



AWS Certified DevOps Engineer - Professional (DOP-C02): Security and Compliance

Identity and Access, and Multi-Account Environments

Implement Techniques for Identity and Access Management at Scale

Hey, Cloud Gurus. Welcome to this lesson, Implement Techniques for Identity and Access Management at Scale. So this domain 6, Security and Compliance, is a new domain. And I decided to just present this in a sort of a different approach. And the title of this lesson is actually a task in domain 6. So let's take a look at that real quick. So we have domain 6 of the exam guide, and there is our lesson title. And it details the areas that we should certainly have knowledge of. So the lesson, we're certainly going to talk about some of these topics. And we'll find that there's a lot of overlap in these domains, and some of these items we've covered already. So some of it should be a little bit of a review. So with all of that said, in the lesson, we'll take a look at a big picture of some of these services and techniques. Then we're going to take a look at policies and permissions, permission boundaries, and we'll finish with some exam tips. So let's take a look at the big picture on Identity and Access Management at scale. And we'll just ask and answer some questions along the way. So what are our best practices for the root account? And this is not an all-encompassing list, but we certainly want to use the root account sparingly. So, first thing we want to do is create admins. And if that isn't the first thing we want to do, well, the first thing would be to set up MFA for our root account. So we have our root user, and we'll create an admin. Our admin seems to be busting a move. They're pretty happy. And we can set up MFA for the root and MFA for the admin as well. Now, what about least privilege? How do we handle that? Well, we could create some groups, attach policies to each group, maybe a network team, a dev team. As we'll see, we have other forms of access such as attribute-based access, but this is still certainly viable. Well, what about users outside of our organization? They need access to our account. We can use identity federation techniques and give them temporary access with least privilege. We can use Simple Token Service to give them temporary credentials, and they get access. All right, so that's a lot of IAM. But how does this scale? Well, setting up Single Sign-On is a good start. And just remember that Single Sign-On has been renamed. It's now called AWS Identity Center. We can get access to our account. We can get access to our on-



premises all with this Single Sign-On. How do we manage all of this across multiple accounts? And you should be thinking AWS Organizations. Just showing four accounts, but there could be many accounts, and it gets complicated at scale. So Organizations is certainly a help. So these look like policies, but for Organizations, they are called service control policies, SCPs. So we're doing IAM at scale now. One last question, where do we store our secrets? So let's say our app needs database access credentials. Of course, we store those in Secrets Manager. There's a lot going on there, but sort of broken down step by step. All pretty straightforward. All right, let's talk about policies and permissions. So we have multiple options here. We have identity-based policies, probably the ones we're most familiar with. We can attach policies to our identities, including and, of course, IAM roles. Resource-based policies, another option for us and certainly on our S3 buckets is probably the most common use of these. Now we have session-based policies. So, we have a user or a role and need to perform a specific task working with a resource. In this case, an S3 bucket. So session policies are advanced policies that you pass as a parameter when you programmatically create a temporary session for user or role. And again, in this case, the policy would contain permissions to work with S3. Now we don't have to choose one or the other, we can use all of these techniques. And if there happens to be a case where these policies are applied, let's say to the same resource, what happens is there's overlap in these policies. And where that overlap occurs defines the effect of permissions. All right, let's talk about permission boundaries for IAM entities, users or roles. Permission boundary is an advanced feature for using a managed policy to set the maximum permissions that an identity-based policy can grant to an IAM entity. So, IAM entity, let's think about users or a role. And here would be a typical permission boundary. So, in the permission boundary, the boundary we're setting is that, hey, we can give access to S3, CloudWatch, and EC2. So with this policy, you could set a permissions boundary that would limit the user or role to permissions only in S3, CloudWatch, and EC2. And of course, the asterisk is indicating pretty much means everything in S3 or CloudWatch or EC2. All right, let's talk about some exam tips. Just a good thing to remember. What is the new name for AWS Single Sign-On? AWS Identity Center. So in a scenario question, if you see the need for Single Sign-On, just remember AWS Identity Center. How can you implement an AWS security best practice for root and admin accounts? Add multi-factor authentication to each account. And specifically for root, sort of lock that away and use it as little as possible. Create those admin accounts and use those. How do you set the maximum permissions that an identity-based policy can grant to an IAM entity? We create a permissions boundary. All right, so that's a look at IAM at scale and also some policies and permissions. So with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

Apply Automation for Security Controls and Data Protection



Hey, Cloud Gurus. Welcome to this lesson, Apply Automation for Security Controls and Data Protection. So this is task 2 for domain 6, Security and Compliance. And because this is a new domain, we're just taking a little deeper dive from the perspective of the DevOps Pro exam guide. Let's take a look at that real quick. And we see task 2, and it requires knowledge of network security components. So, some basic networking knowledge with the addition of Network Firewall, AWS WAF, and Shield, certificates and public key infrastructure, some data management. Some of the services include Security Hub, Organizations, Control Tower, Systems Manager, again, WAF, Config, which is closely coupled with Security Hub. And some of these services we've already had lessons on. So let's take a look at task 2, Apply Automation for Security Controls and Data Protection. So in this lesson, we'll start by looking at AWS Network Firewall, and we'll take a look at overall network security, combined security posture, and security at scale, multi account, multi region. And we'll finish up with some exam tips. So let's look at Network Firewall, and Network Firewall is a stateful managed Network Firewall intrusion, detection, and prevention service for our VPCs. Now, what we're doing with Network Firewall is filtering traffic. All right, so we have traffic coming into our internet gateway and also internal traffic going out through our internet gateway. And what we're going to do with Network Firewall, so we create a Network Firewall subnet. Within that subnet, we have a firewall endpoint. We insert this between our internet gateway and our customer subnet, of course, subnets. So any traffic coming in through our gateway, before it hits our subnets, goes through this firewall, and it's filtered. And of course, we can set up rules on how we want to filter the traffic, and we can filter out known bad domains and such things. And again, this is two-way. So, we're filtering traffic coming in, but we can also filter going out. An example would be we could disallow users in our subnets to view certain sites. We could filter out. We could disallow it right here, the Network Firewall. So it filters traffic going between the subnets and locations outside of your VPC. And it's just important to understand that it's two-way traffic that it can filter. How do we do this? Well, the key to it is we modify our route tables to send network traffic through the network firewall's endpoints. So, we have a route table for our internet gateway, and we change the route that previously was going to the customer subnet. We change that route to go to this Network Firewall. Likewise, the subnet has a route table. And rather than going straight to the internet gateway, we alter that so that it goes straight to the Network Firewall. And of course, the Network Firewall routes need to finish that delivery or, of course, filter out the traffic that is unwanted. So that's a higher-level look at AWS Network Firewall. We'll keep it high level because we have a hands-on lab on Network Firewall and certainly solidify all of these concepts. All right, let's take a look at some network security concepts. How do we protect against attacks and unauthorized access? So, it's key to make the difference between these two, attacks and unauthorized access. What services handle each of those? So, AWS WAF operates at layer 7 of the OSI model. So just think of the name, the literal name, WAF, Web Application Firewall. So it's a network security firewall that protects web applications from HTTP and HTTPS



and web app security vulnerabilities. And think of attacks such as SQL injection, cross-site scripting, and DDoS attacks. Of course, we have Network Firewall as well, and this operates at layers 3 and 4 of the OSI model. Now here's your difference. WAF is protecting against those attacks. Network Firewall is filtering unauthorized access. So just understand that that difference and also the layers that they operate. Now, we have a third player here, AWS Shield. Shield is a managed distributed denial of service, DDoS, protection service that safeguards applications running on AWS. Shield also operates at layers 3 and 4. So if we get a web attack, WAF can block that. Unauthorized access, Network Firewall. Shield is focused on those DDoS attacks, although WAF can also block the DDoS attacks. So if you understand those key concepts, you'll be fine. All right, let's talk about combined security. And just remember, WAF Firewall and Shield can still be in play here. Just for the sake of space, let's just visualize in our mind that we have WAF firewall and Shield set up. So some other services that contribute are— Well, at our load balancer, in conjunction with AWS Certificate Manager, we can have certificates on our load balancer So we can associate our certificates with the load balancer. And traffic coming in is going to need to programmatically request those certificates, well request a certificate, or it's just not going to go any further. So with these certificates provided and managed by Certificate Manager, we can authenticate and encrypt our internet traffic. And speaking of combined security, well, let's talk about AWS Config and its partner in fighting crime, Security Hub. So, a prerequisite for enabling Security Hub is that you have to enable AWS Config first. That's how reliant upon Config that Security Hub is. And with Security Hub, we can start receiving findings on security vulnerabilities in our accounts. So, Security Hub uses Config rules to perform security checks. And what are some of these common security checks? Is MFA enabled on your root account? And that will be a critical finding in Security Hub. Are you prohibiting public read on your S3 buckets Is access restricted on security groups? And specifically, are you using IP address ranges instead of just everyone 0.0.0.0/0? Those will be Security Hub findings as well. All right, let's talk about security at scale a little bit, multi account and multi region. All right. So we have multiple accounts in multiple regions. Now, Security Hub can operate in this environment. We have to designate an aggregation region. So, that's where all of our findings will aggregate. And then other regions can send findings to that aggregation region. Other things we can do, Systems Manager and Run Command. We can run automations across multiple AWS regions, AWS accounts, and/or organizations. You'll find that organizations is very worthwhile to set up because it just helps so much in securing our accounts at the enterprise level. All right, so let's assume we have Organizations set up and organizational units, the OUs. Remember, Control Tower is closely linked with Organizations and the OUs. With Control Tower, we can configure a central location for governance and security. You scroll up a little bit, and that is the landing zone. When we create a landing zone, if we have not configured Organizations, it will be done for us. Use organizational OUs to set up a landing zone. So if we don't do this, it will be done for us to set up this landing zone. So the landing zone relies on these



organizational units. All right, so that's a bit of security at scale, multi account, multi region. With all that said, let's finish up with some exam tips. With Network Firewall, what facilitates the connection between the internet gateway and the customer subnet? It's going to be that firewall endpoint. And again, we'll see this in the hands-on lab. Which layers do WAF and Network Firewall operate, respectively? Key point here, WAF is at layer 7 Network Firewall is at layers 3 and 4 as is Shield at 3 and 4. How do AWS Organizations OUs help Control Tower? They can be used to help set up the Control Tower landing zone. If you don't have those OUs when the landing zone is being created, those OUs will be created for you. And with that, I'm going to end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

Implement Security Monitoring and Auditing Solutions

Hey, Cloud Gurus. Welcome to this lesson, Implement security Monitoring and Auditing Solutions. Now this is another domain 6 task from the AWS DevOps exam guide. And being that this is a new domain, I decided to look at it, really take a deep dive into it, sort of an itemized walk through domain 6. So that said, let's take a quick peek at the exam guide. And here we are, task statement 3, Implement Security Monitoring and Auditing Solutions. So within this task, I have to have a knowledge of security auditing services, CloudTrail, AWS Config, VPC Flow Logs, drift detection. Some of the related services, GuardDuty Inspector, the relatively new IAM Access Analyzer, AWS Config, and a knowledge of common cloud security threats, insecure web traffic, exposed access keys, S3 buckets with public access enabled. Now for this bullet point specifically, if you have Security Hub enabled, it will point a lot of this out to you. You can actually learn from Security Hub and going forward be proactive. So with that said, in this lesson, we'll talk about security auditing, identifying vulnerabilities, and then we'll do a demo, looking at some security vulnerabilities and also IAM Access Analyzer. And we'll finish, as always, with some exam tips. So let's get started and take a look at security auditing. All right. Now, when you see the term auditing with regard to AWS, definitely want to start thinking about AWS Config. And how do we interact with AWS Config? Well, AWS Config stores log files in S3. But of course, we can review changes from the console. AWS Config is integrated with CloudWatch Metrics. And being integrated with CloudWatch metrics, AWS Config can capture several usage metrics related to the resources in our environment. And it's another way, again, that we have sort of a window into AWS Config. And as always, we can automate and send messages via SNS. So that's Config. Now remember that AWS Config also has automatic remediation. And sort of linking terms with services, when you think of automatic remediation, it's good to start thinking about Systems Manager automations. All right, so CloudFormation stack. How do we audit and maintain the integrity of our stacks? And we've had plenty on this throughout the course. CloudFormation drift detection, and we can go in and manually correct our drift, or we can set up



automatic remediation. What about VPC Flow Logs? Those flow logs go to S3, and we can publish those flow logs to CloudWatch Logs. We can also set up Kinesis Data Firehose to receive these logs and send them downstream for further analysis. And we have CloudTrail. CloudTrail log files can be stored in S3. So, obviously S3 is an accumulation point for our log files. CloudWatch Logs can be sent to S3 as well. So we're sending all of these log files into S3. And one way we can analyze these log files is with Amazon QuickSight. QuickSight has some machine learning capabilities and analysis, graphing. Also, we can do queries with Amazon Athena. I'm showing an error to the VPC Flow Logs. Of course, those flow logs can be stored in S3 as well. So that's sort of around-the-horn look at security auditing, some of the services we can use and how we can capture those log files. S3 plays a big part in this and certainly some of our analysis tools, QuickSight and Athena. Think about moving large bits of data around, something like Firehose. So, let's take a look at identifying vulnerabilities and just have sort of a question and answer session. So we need to scan AWS workloads for software vulnerabilities and unintended network exposure. And this is another case where if we simply know what the service does for us, it'll serve us very well on the exam. And that service is Amazon Inspector. Now what if we need to monitor our AWS accounts and workloads for malicious activity? Sort of match the term with the service. That service is GuardDuty. And certainly, knowing the difference between the two will serve you well. So what is the difference? Inspector works more at the application level. Think about software vulnerabilities on applications. GuardDuty works at the overall account level. All right. What if we need to identify resources in our organization and accounts that are shared with an external entity? So who's coming in from the outside? What kind of access do they have? Are there security vulnerabilities? The service for that is IAM Access Analyzer, and there's a link here that provides all the services Access Analyzer works with. And you can go out and check that out, but I'll just rattle off the prominent ones here, S3 buckets, IAM roles, Key Management Service, Lambda Functions, SQS, Secrets Manager, SNS, Elastic Block Store, Snapshots, RDS Snapshots, ECR repositories, and EFS file systems, which is actually kind of redundant, Elastic File System. All right, and we're going to check out Access Analyzer in the demo. Some rapid fire questions. Are your access keys rotated? Is WAF enabled? Are you using hardened AMIs? Do you have S3 buckets with public read? So that's a great list. And if you're answering properly, you're a long way toward securing your account. There are Config rules for all of these potential vulnerabilities, and that's pretty impressive. And remember, with Config enabled, we can enable Security Hub and consolidate findings there. We'll take a peek at Security Hub in the demo as well. And with that said, let's go out to the management console and check out some security vulnerabilities and the IAM Access Analyzer. All right, out at the management console, let's go over to IAM. All right and I have some warnings here. Add MFA for the root user. That's certainly one that will produce a finding in Security Hub. What about access keys? Don't keep your access keys hanging around. So, I'm not going to go the whole way through with this just for time, but I would deactivate, then delete those access



keys. Let's go over to EC2. And I'm not going to launch an instance, but let's start. And note what's going on here by default, SSH traffic from anywhere. Now, it does provide a warning. But guess what? This will produce a finding in Security Hub. So it would be better to provide an IP address range for appropriate access to SSH. And I'll just mention, be careful with that and security groups as well and how you implement your resources in your AMIs and CloudFormation templates. Let's go over to Security Hub real quick. And in addition to being great for identifying security vulnerabilities, there's that MFA, pesky MFA, root access keys, public read on S3 buckets. CloudFront distributions should have a default root object. So, in this way, you're directing where the end user is going to land on your CloudFront and just not have free rein. Security groups should not allow unrestricted access. So, with that said, we can learn from Security Hub and be proactive and take care of these things up front. All right, let's check out IAM Access Analyzer. And actually, I'll just close this. We're at IAM already on the navigation pane. Here's Access Analyzer. How it works, we create an analyzer, review the active findings, and take action. And we ultimately can archive findings. So, let's create an analyzer, and this just takes a couple of minutes. And it's using— I have AWS Organizations configured. It's using my organization. Trust relationship is created between Organizations and IAM a Access Analyzer. Service-linked role is created in all accounts in the organization, which gives Access Analyzer permission to interact with resources on your behalf. So let's create. All right. And this just takes a couple minutes to produce some findings. All right, analyzer creation is complete. Takes another minute or two to actually populate some findings. So while that is going on, Access Analyzer makes it easier to implement least privileged permissions by generating IAM policies based on access activity. Probably don't need to go that far with it, but let's talk about Access Analyzer. What it does, helps you identify the resources in your organization and accounts, such as S3 buckets or IAM roles shared with an external entity who has access to our account from the outside. And this lets us identify unintended accesses to our resources and data. And that of course, is a security risk. All right. So we have some findings and resource S3 bucket IAM roles, and this is the account number which I'm not showing the whole account number. But it's saying this bucket's wide open, and there's a cross-account role going on with these IAM roles. Now, that may be fine, but it's just telling us what's going on. We may have created a cross-account role. The external person did what they need to do. They don't need access anymore, least privilege. So we could get rid of these. And certainly, for the S3 buckets, we want to tighten those buckets up. They probably have read access on them. So, we can go about resolving these, and then we can archive them. So that one got archived. So that, in a nutshell, is what IAM Access Analyzer does for us. All right, let's check out some exam tips. What are the key services that can analyze? CloudTrail Log files, QuickSight, Athena, and CloudWatch. What can you use to set up automatic notification of configuration changes? AWS Config with SNS and certainly with Lambda. If you chose to work with Lambda, a common pattern is for Lambda to send information to a Slack channel. And finally, how can you identify resources



in your account that can be accessed externally? So our short-term memory is probably good enough that we know that is IAM Access Analyzer. All right, with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

Role and Attribute Policies

Hey, Cloud Gurus. Welcome to this lesson, Role and Attribute Policies. So in this lesson, we'll take a quick look at role-based access, which we certainly know a lot about at this point. And then we'll take a look at the relatively new attribute-based access. So we'll start off talking about the types of access. And again, the role-based access will be a bit of a review. And then we'll go deeper on attribute-based access. That acronym ABAC, attribute-based access. Key point of ABAC is tagging. So we'll talk about tagging. And then, to solidify these thoughts, we'll do a demo on attribute-based access, and we'll finish up with some exam tips. So, let's start talking about these types of access. So again, we're very familiar with role-based access. We have an IAM role, which of course has a policy attached to it, defining the permissions. And with this role given to an identity, we can access AWS resources. So of course, we can attach this role to a user. We can attach the role to a pipeline deployment. But what are the drawbacks? Well, let's say we have a large organization. We can have unique users, many of them, with multiple roles. So it can get a bit messy at scale. And we can have complex logic with difficult-to-define roles. Now add all of this together, again, let's talk about large organizations, many users, maybe many accounts, several regions. This gets complex, scaling up. And where it becomes costly is, from a time perspective, there's constant administration. And authorizations are based on dynamic parameters constantly changing. Just a constantly changing playing field. And again, constant administration. Now, if you spend any amount of time in AWS, you've certainly used role-based access. Smaller scale works beautifully. No problem at all. At scale, it can become complicated. So let's talk about attribute-based access. Been around a few years, but certainly newer than role-based access. So what does this mean? Well, it determines access to resources based on attributes. These attributes can be associated with a user, a resource, environment, or even an application state. So, we determine access to resources based on attributes, and we associate them, the user or resource. And we'll see how that plays out here momentarily. The advantage is this ABAC model is very flexible in allowing dynamic, contextual, and granular authorization decisions. Now, it sounds complicated. It's a little bit more difficult to deploy initially. Let's go a little bit further. So our identities, users, groups, roles, are resources, S3 bucket, maybe some EC2 instances, RDS. And from a high-level perspective, if an identity— And let's be specific here. Let's say we have a user and a resource, an S3 bucket. If an identity and a resource share an attribute, then the identity can access that resource. So how do we set up the attributes is the question there. We do that with tagging. So, tags are key pairs just like we could



add tags to EC2 instances or any of our resources in AWS. There's a tag key and a tag value. This becomes very versatile. Let's say that we give our key the name Environment. Our tag values could be Dev, Test, Prod. So let's see this in action. All right. So we have a tag, Environment, with three possible values, Dev, Test, and Prod. I want to keep it fairly simple here. But what if this tag was account? And what if we had 20 accounts? So we could have 20 possible values. Then you start to see the scalability of this pattern. So we have developers, and we want our developers to be able to access the dev and the test environments. Well, simple enough. They need these tags, Dev and Test. These servers over here have these tags as well. Developers can access them. What about admins? Admin has access to everything, Dev, Test, Prod. You want minimal access to prod, maybe a couple administrators. And how about our pipeline account? Other than the admin, our pipeline account is the only entity to access Prod. So we just give it that tag. So with one tag, we created all of these different types of access. And again, maybe the tag could be account with 20 accounts. This access to the accounts, it could look complicated. But really, it's just a matter of attaching the right tags to each of the entities. All of this varying access across multiple users accomplished with a single tag, limited to three possible choices. But again, this could be much larger possible choices. So with that foundational knowledge on attribute-based access, let's see how this looks with a quick demo. All right, for our demo, we'll take a look at a few policies which will illustrate the benefits of attribute-based access in certain use cases. So, to set this up, let's say that we have 20 teams. We're in a large organization, and these teams need to manage their secrets and their databases. None of these teams have access to any of the resources of the other teams. They just need their own resources and to manage those resources. So, using the resource-based approach, our administrator creates a policy for each team. Of course, we can attach that policy to the appropriate users or create groups and attach the policy to the groups. I'm just showing two policies. Again, the premise is we have 20 teams. Each team would have their own policy. So, in the resource-based policy, this first statements just allowing the user to view the secrets in the console. And this policy is giving them access to their secrets. And note that for this other policy, these secrets are different. They're specific to each team. Now remember the premise that we have 20 teams. So, what if a team needs a new secret? Well, our administrator has to update this policy, and that can happen for 20 teams. So this approach isn't scaling well. Let's take a look at an attribute-based policy. So, again, this is just allowing the team members to view the secret. And with that, we can use tagging. So, each team, we create a tag. And in this case, it's called groupid. So we create that tag. So we have 20 teams. Let's say 1 through 20. Groupid for each team correspondingly 1 through 20. So our teams and team members in each group have that groupid tag. Now, with resource-based policies, we had to have a policy for each team. This policy can be used by all teams because of that tagging. So let's go through this policy real quick. Again, this ListSecrets, that's for viewing the secrets in the management console. The second one, create a secret. We're allowing our users to create a secret with the condition that they



tag it properly. They have a tag. This secret has to have a matching tag. Now, this statement is giving them permission to manage all of their secrets, again, with this stipulation, the condition that their personal tag matches the resource. So if the secret has that same groupid as the user, they can manage that secret. Now how do we protect this? Well, we disallow untagging resources. So in our premise, with resource-based access, we would have had to have a policy for each team. Our administrator has to update policy when a new secret is needed. With attribute-based, one policy for all of those teams. The thing that is going to change is our tag. So that tag can be applied to 20 teams or even more. And now the users can create their own secrets, manage their own secrets, and it's protected from untagging resources. So that's a look at a use case where attribute-based access really shines. And it can be really beneficial in large organizations where our permissions need to scale. All right, let's talk about some exam tips for attribute-based access. How are conditions used in attribute-based access? Well, just think of it as an if statement. If an identity, a user, a group, a role, or a resource, shares an attribute, a tag, with a resource, then the identity can access that resource. Kind of already answered this one. How are attributes assigned in ABAC? Using tags. Why are tags so versatile? We saw that in the lesson and the demo. It's a key pair, and it can have multiple values. That one tag, we saw an instance where it had three values, but could many more values and become very powerful as our environment scales. So that's a look at resource and attribute-based access. With that, I'll end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

Identity Providers and Federation

Hey, Cloud Gurus. Welcome to this lesson, Identity Providers and Federation. So in this lesson, we're going to talk about identity providers, identity federation. We will talk about AWS Identity Center, and we'll see what that's all about. Go ahead and do a quick demo on Identity Center, and we will wrap up with some exam tips. So let's go ahead and talk about identity providers. Now what are we talking about? Well, external access to our AWS account or accounts. So users outside of our organization who need access, maybe they are subcontractors. An identity provider allows external users to log in, here's the key, without creating IAM users for them. It eliminates a possible administration nightmare and maybe a security issue where we would have unused accounts hanging around. Possible identity providers, and this is by no means a complete list, but ones we are certainly familiar with and probably familiar with as identity providers, Amazon, Facebook, Google. We do this all the time. If we're using the the internet, get the option do you want to log in with your Google account or Amazon, log in using Facebook. That's what's going on. We set up a trust relationship. And because of that trust, users can use these accounts to get access to our AWS account. So, we must create a trust between the account and the identity provider. Now, these providers must be compatible with either



OpenID Connect or SAML 2.0. All right. So we know what identity providers are. Let's talk more about identity federation, specific types of identity providers, web identity federation and SAML 2.0 federation. Now key points regarding web identity federation. Use cases. Let's say we have a mobile app and a game on a mobile app, and our users need access to store their results in DynamoDB. That would be a typical use case. How do we safely provide temporary AWS credentials? Web identity federation. Let's see this in action a little bit. So we have our identity providers. Our users can authenticate. We probably get asked this question a couple times a day. Do we want to use our Gmail account to log in? We can say yes. Because of that trust that's set up, we get authorization from web identity federation. And what happens is Security Token Service will give us a temporary token for which we can then log in to the AWS account. Now what about multiple account environments? And maybe there's an on-prem account. Things get complicated. How do we handle that? AWS Identity Center. Now this might be a new term to you, but it's basically the successor to AWS Single Sign-On. So again, let's say we have several accounts. Maybe we have on-prem servers. Identity Center now, again, the successor to Single Sign-On, we can set it up so that one login can get us access to all these different accounts. All right, let's go a little bit deeper. Key components of AWS Identity Center workforce identities. Used to centrally manage workforce users and groups. So, human users who are members of your organization are known as workforce identities or workforce users, and they can be created in Identity Center. Or you can connect and synchronize to an existing set of users and groups in your own identity source. Let's talk on-prem. So, we could have Microsoft Active Directory on-prem. We can synchronize with that. Or we could synchronize with Okta Universal Directory or Microsoft Azure AD. So, the workforce identities, long story short, are your users. Where are they coming from? Maybe we need to synchronize. We also have multi account permissions. It's used to centrally manage AWS account access. Here's where we implement our IAM permissions across multiple AWS accounts at one time without needing to configure each of those accounts manually. So that is very powerful. Another part of Identity Center, application assignments. Used to centrally manage access to our AWS applications. What types of applications? Well, how about Salesforce, Microsoft 365. With application assignments, we can grant our workforce identities access to these apps. So using Single Sign-On, these could be SAML applications, again, such as Salesforce and Microsoft 365. So Identity Center is a pretty deep topic. Just good to have a general understanding of what it does. Certainly remember that it is the successor to Single Sign-On. Now you may see the term Single Sign-On as a generic general term. Just remember, AWS Identity Center provides that capability for us. Let's take a look at Identity Center from the management console to solidify these topics a bit. All right, at the management console, Identity Center, the successor to Single Sign-On, Enable. Now a key point is that Organizations, AWS Organizations, is tightly coupled to Identity Center. We must have Organizations running. If we don't have it set up, Identity



Center will do it for us. So let's enable this. Step 1, we choose our identity source. This is where we administer users and groups, and it's the service that authenticates your users. Now in step 2, we have multiple options here. Manage access to multiple accounts, set up Identity Center-enabled applications. Going back to that example, something like Salesforce or Microsoft 365. Manage assignments to your cloud applications. Think about Microsoft AD or Azure Active Directory or Okta. Let's go ahead and choose our identity source. And the identity source is Identity Center Directory. So we just enabled Identity Center. And behind the scenes, it automatically configured this Identity Center directory for us as our default identity source. So this is where we create our users and groups and assign their access levels. Now what we could do, go over to Actions, we could change the identity source. So, we had on-prem Microsoft AD. We could select that. Or maybe we're using Okta external identity provider. But let's just keep it simple here and go with the default. We can set up our authentication. Configure standard authentication. By default, users created do not have a password. We can send an email. And we could set up multifactor authentication and the situations where we want to use MFA. How they can authenticate, examples being Authy, Google Authenticator, Microsoft Authenticator. Let's go to Users. We could start adding users. And now let's click Add user first. We can start entering users and their details. Certainly, the email address is important. Send them their password. We can also find users. Once we have users, we could add some groups, create groups, and then add our users to the appropriate group. All right, don't want this demo to get too long. Just wanted to give you sort of a flavor of what Identity Center can do for us. We can consolidate into Single Sign-On over multiple accounts on-prem, Microsoft AD, or external, things like Okta with the primary goal of having our users be able to log in to all of these different accounts in one place and also to be able to manage them in one place. All right, let's talk about some exam tips. Which identity providers can be used to log in into AWS? Now this isn't a complete list, but probably see this once or twice a day, at least. Amazon, Google, Facebook get asked the question, do you want to log in with your Gmail account? That's what's going on here. An identity provider must be compatible with which formats? OpenID Connect or SAML 2.0. And most likely in a scenario question, you might see one of these, and you'll know to be thinking about web identity federation. What service provides temporary tokens for AWS access? That is AWS security Token Service, otherwise known as AWS STS. So let's a look at identity providers, identity federation, and AWS Identity Center, the successor to AWS Single Sign-On. With that, I will end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

AWS Organizations for Multi-Account Environments

Hey, Cloud Gurus. Welcome. In this lesson, we're going to talk about AWS Organizations for multi account environments. So what is AWS



Organizations? Well, it's an account management service that enables you to consolidate multiple AWS accounts into an organization you create and centrally manage. Think of a large company with multiple offices that has multiple AWS accounts, which spans multiple AWS regions. So how do you manage all of that? The answer is AWS Organizations. Organizations includes account management consolidated billing capabilities, and this enables us to better meet our budgetary, security, and compliance needs. So, in this lesson, we'll do a little bit more of an overview, talk about the requirements and use cases, talk about control. And in that instance, we're talking about security controls. And then we'll just move on to some exam tips. So with that, let's talk about AWS Organizations. Now, with AWS Organizations, you're going to have to have one management account. And from that management account, we can create our AWS Organizations. And what we have to do is invite other accounts to join the organization. Now it's probably not necessary for the exam to go through this organization creation process. But if you're interested, I do have a supplemental lesson, which I'll include in the lesson materials, in which I actually create an organization. I invite a second account and configure the organization from there. So we have our management account. And again, the organization is a collection of AWS accounts, those accounts which we invite and they have to accept the invitation, and then they are part of the organization. And we can organize these accounts into a hierarchy and manage them centrally. And we have what are called organizational units. And that is a group of AWS accounts within an organization. How would we determine what belongs in an organizational unit? Well, it could be a hierarchy such as dev, test, and prod groups. It could be by department. It could be by office location, maybe a geographically dispersed organization. So those are sort of the categories that we would have for organizational units. Now, in this diagram, I'm showing an organizational unit, and notice sort of this tree structure. Under the organizational unit, we can have other accounts, we can have sort of a nesting another organizational unit as well. Now, speaking of accounts again, remember we have our management account who created the organization and then all the other accounts are known as member accounts. Any account that isn't the management account is a member account. Now anytime we're considering using an AWS service, we have to think about, hey, what's the cost of this? What about billing? Who pays for all of this. So, there's no additional charge for organization, but there's always a catch. There's no free lunch. The owner of the management account is responsible for paying for all usage, data, and resources used by the accounts in the organization. And that sort of makes sense. That's why we're creating the organization. We want to manage everything in a central location, and that includes billing. Now that's not viable in an organization. Certainly, something needs to be considered upfront. All right, let's talk a little bit about requirements and also use cases really. So when do we use AWS Organizations? So let's say we have a large organization with multiple AWS accounts. And remember, again, this can span AWS regions. So we want to implement and enforce corporate security, audit, and compliance policies across



all of our accounts. We create an organization and we can manage that centrally. Also, we need to share resources across accounts. Same situation. It's much easier when we create an AWS Organization. What about automating the creation of AWS accounts and categorize workloads using groups? So in Organizations, we can actually create new AWS accounts. Now getting more specific, think about CloudFormation stack sets. When do we use those? When we want to deploy stacks across AWS regions and with multiple accounts. So stack sets fit perfectly with AWS Organizations. And in an upcoming lesson, we're going to be talking about AWS Service Catalog where we are distributing applications and resources across our organization. And yes, Service Catalog also works great with Organizations. Other services that can span our organization include CloudTrail, EventBridge events, and AWS Config, to name a few. But those are sort of the really important ones that contribute to organizational management. So if you run into a scenario where you need to manage CloudFormation stack sets centrally in one location, that answer is AWS Organizations. You cannot do that with IAM. All right, let's talk a little bit about control, and I'm speaking from a security perspective. So we have our management account, AWS Organizations, and we have some setups here. Project A, this is an OU, organizational unit, and this is spelled out specifically or generically maybe. We've seen this hierarchy before. Our OUs can be nested. Another way to do. This is sort of specific OUs for functional teams, prod staging, dev. But how do we control all of this? We do it with policies. Now, we do this with regular IAM policies as you would expect. But additionally, Organizations have what are called SCPs, service control policy. These define the AWS service actions such as Amazon EC2 run instances that are available for use in different accounts within the organization. Now it's just important to understand the difference between SCPs and regular policies. Now we know what regular policies are, so I'm just going to speak to SCPs. SCPs are a type of organizational policy. They're specific to AWS Organizations we use to manage permissions in our organization. Now, here's the catch. SCPs alone are not sufficient to granting permissions to the account in our organization. No permissions are granted by an SCP. SCPs are sort of a guardrail, sort of they set limits on the actions that our account administrator can delegate to regular users. But we still have to refine the permissions. We still have to attach our regular identity-based or resource policy to our users or roles or resources in the accounts. So the SCP set the guardrails for what we can do in the organization, and then it's refined. And the actual permissions are granted by our regular policies that we see everywhere in AWS. So that is a look at AWS Organizations for multi-account environments. Let's talk about some exam tips. All right, use cases for Organizations. Automate the creation of AWS accounts and categorize workloads using groups. The AWS accounts can actually be created within Organizations. Implement and enforce audit and compliance policies. And remember, AWS Config can work with organization. Provide tools and access for your security teams while encouraging development. So, basically, it's bringing your organizations together, your accounts together, maybe in a widespread, multiregional environment. And we



can share common resources across accounts and services that really, really are well suited to work within Organizations, CloudFormation stack sets and AWS Service Catalog. And in a lesson in this section, we will have a lesson on AWS Service Catalog. So think of it in the context of working with Organizations. So that's a look at AWS Organizations, and I'm going to end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Multi-Account Scenarios with AWS Organizations

Hey, Cloud Gurus. Welcome to this lesson, Multi-Account scenarios with AWS Organizations. So this new version of the DevOps Pro exam, DOP-C02, has a heavy focus on multi-account and multi-region scenarios. And a big driver behind that is the service AWS Organizations. So we're going to take another look at AWS Organizations from that perspective. So in this lesson, we'll look at setting up trusted access in AWS Organizations and just general security, design principles, monitoring. And we'll see how AWS Config works with Organizations across multiple accounts. And we'll wrap up with some scenarios and exam tips. So with all that said, let's start off by talking about trusted access. All right, so we've set up AWS Organizations, and we have all of our accounts. And we need some services to work with our Organizations. How do we do that? Well, we set up trusted access. We can use trusted access to enable a supported AWS service called the trusted service to perform tasks in our organization and its accounts on our behalf. A great analogy to make here is think of a service role. A service role is performing actions on our behalf. Well, this scales it up an order of magnitude. Trusted access can enable services to perform actions on our behalf at the AWS Organizations scale. That's a good way to look at it. So, say we have a trusted service, and a good example is CloudFormation stack sets. Well, think about what stack sets do. It deploys stacks across multiple accounts and multiple regions. And that sort of just screams an interaction with AWS Organizations. How do we do that? All we have to do is set up this trusted access, and we'll take a look at that momentarily. But once stack sets has trusted access, it can go ahead and deploy to multiple accounts in multiple regions. And let's take a look at some key services that Organizations supports for trusted access. CloudTrail. So we can have CloudTrail trails across our multiple accounts. AWS Config, and the AWS Config aggregator helps with that. We'll talk about that in the lesson. Control Tower, and Control Tower is really tightly coupled with Organizations. So that's probably no surprise. Single Sign-On, which the service itself is now known as Identity Center. That certainly makes sense as well, Single Sign-On across multiple accounts. And Systems Manager. And thinking some of the things Systems Manager can do such as run automations, and this will enable it across multiple accounts. Now those are some of the key services for which we can set up trusted access. But let's go to the management console and just take a quick look at where we set up trusted access. And in doing so, we'll see all of the services for which we can set up trusted access. All right, so I'm logged into the



management console, and let's just go over to AWS Organizations. And let's click on Services. All right. So, we can configure supported AWS services to perform actions in our organization. And the key to this is there's our trusted access. And we need to enable trusted access. So, here's the list. Amazon Detective, DevOps Guru, GuardDuty, Inspector, Macie, a lot of security. And I'll just scan down through this. There's Control Tower. Note, access already enabled. Single Sign-On. I've already enabled access. CloudTrail, AWS Config. I'm showing all services. We can break those up by categories. But just for the purposes of the demo, let's say that we want to enable Security Hub. So I click on it, and it is just this simple. Click Enable trusted access. So it's recommending we enable trusted access by using the Security Hub console. And I'm just going to do it here. Just wanted to just emphasize the general idea of the need to set up trusted access and also show the services. So that's how we do it, and I really like the analogy of comparing that to a service role on the larger scale, multi-account and multi-region, with the driver being AWS Organizations. All right, let's talk a little bit about security. All right, so we have our organization. And with AWS Organizations, we have service control policies. The service control policy defines the boundaries that can be accessed in an organizational unit or member account. So, the SCP sets up a permission boundary. We still have to work with policies, but we can't give any permissions outside the bounds of the service control policy. We could try; it just won't work. The SCP can be attached to organizational units or member accounts. If the SCP denies an action on an account, no entity in the account can perform that action, even if its IAM permissions allow it. So it's a combination of those two that actually gives the permissions. And it's important to understand the role the SCP plays. So we have our SCP. Ah, remember, IAM permission boundaries as well. They may or may not exist. But if they do, they contribute, and then we have the role permissions. And ultimately, where all of these meet in the middle is where the permissions are defined. But again, that permission boundary of the SCP. So we wouldn't have any permissions outside of the permissions boundary. No matter how hard we try, just won't work. Let's talk about some key design principles for Organizations. And one key is we want to use organizational units based on function rather than mirroring your organization's reporting structure. Another way to say that is by function, not by department necessarily. Apply service control policies to organizational units instead of accounts. And these two kind of go hand in hand. If you're following this design principle, then it makes sense to apply SCPs to OUs. Avoid deep hierarchies. Sure, we can do three or four levels here, but it's messy and error-prone. So maybe just two levels of our organizational units. And this is common good practice. Separate production and non production workloads. Use federated access, IAM Identity Center or third party, to avoid having to manage individual users in each account and avoid deploying workloads to the management account. Let the management account manage. All right. What about monitoring in a multi-account environment? Well, we can create a CloudTrail trail that captures all API calls in AWS Organizations. And we can work with the EventBridge. EventBridge event is triggered by API calls



which, of course, CloudTrail captures. The event passes details of the calls to a Lambda function, and it can pass them to an SNS topic as well. The Lambda function can log the event details to an S3 bucket. And then from there, we can do whatever we need to, analysis or whatever. So, it's a common automation chain here. Remember, what's the key? We can set up trusted access. All right. What about AWS Config? And we're always thinking about compliance when talking about Config. Multi-account, multi-region data aggregation. And Config allows you to aggregate data from multiple accounts and regions into a single AWS Config account, and we have that AWS Config aggregator. And the aggregator can collect all of that data for us. Again, I'll repeat. Nothing happens without going into Organizations and setting up that trusted access for AWS Config. All right, what about scenarios? How can we integrate GuardDuty with Organizations to protect our accounts? Well, back to that repeating myself, but we Config GuardDuty in every account. Configure one account as the GuardDuty administrator. Configure EventBridge to send all GuardDuty findings to Kinesis Data Firehose. We can stream those findings to an S3 bucket. Again, first step, set up trusted access for GuardDuty. So these common patterns would work across Organizations. We just have to set it up properly. All right, some key points. What are the services that interact with Organizations? And you can see those if you go into Services in the management console and look at trusted access. Which ones are enabled, which ones you want to enable. Common automation and messaging patterns will work across accounts if AWS Organizations is configured properly, and we start with that trusted access. SCPs are meant to be guardrails. They don't define permissions by themselves, so they do define the boundaries of those permissions. And the harder you try you can't go outside of those boundaries, even if you specifically give permissions in an IAM policy. And along that theme, the SCPs, user permissions, and if you're using permission boundaries, IAM permission boundaries, they all combine to define those permissions. Real quick exam tips. How can you enable an AWS service to work with Organizations. Well, we set up trusted access. Do service control policies define permissions for users and AWS Organizations? Certainly not by themselves. They are boundaries for access, but the policies are still needed. And what is that AWS Config feature? It enables it to work with organizations across multiple constants and regions. It is the aggregator. So again, for the exam, multi-account and multi-region is a big focus, and the key player is AWS Organizations. Really know the fundamentals of working with Organizations, and you'll be in good shape. And with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Systems Manager

Systems Manager Introduction and Setup - Demo



Hey, Cloud Gurus. Welcome. So in this lesson, we're going to look at Systems Manager. We'll go over an introduction of Systems Manager. And we'll go into the management console and do some setup. So we'll take a look at an introductory look at Systems Manager. Talk a bit about resource groups a bit more. Talk about the capabilities that we have in Systems Manager. We'll do a quick demo just to see Systems Manager in action from the management console. Come back and talk about some organizational controls. And we'll finish up with some exam tips. So what is Systems Manager? Well, Systems Manager is a collection of capabilities for managing our virtual machines both on-premises and our EC2 instances in the cloud. So let's talk more about Systems Manager. So the Systems Manager allows us to manage and operate our resources at scale, create groups of resources across different AWS services as above. We can group them and manage them together. We can view aggregated data by resource group. And we can respond to insights and automate operational actions. Remember, in a previous lesson on Security Hub, we talked about remediating security issues on our instances. Well, we do that with Systems Manager. And we've talked about on-premises servers, as well as EC2 instances. How do we manage these? Well, I like to make a parallel when possible. Remember CloudWatch Logs and the CloudWatch Logs Agent. Well, likewise, with Systems Manager, if we want to manage an instance, we install Systems Manager Agent on that instance, whether it's in the cloud or on-premises. All right, now we know a bit about resource groups and tags. So, how do we create resource groups? Well, with resource tags. After tagging resources, we can view a consolidated dashboard in Systems Manager. And Systems Manager comprises the following capability types, operations management, application management, change management, node management, and shared resources. Now, this will make a little bit more sense momentarily when we see these in the management console. These are actually headings, and we'll see the operations under each of these. So Systems Manager might be a new concept for some of us. So let's go out to the management console and tie this all together with a little demo. All right out at the management console and Systems Manager. Now remember our capability types, operations management, application management, change management, node management. Then we have some shared resources. This is how it works on the main page. It's simplified, but it really captures what we're trying to do. We group our resources. Good example, being a group of EC2 instances. View insights. See relevant operational data and dashboards about our grouped resources, and we take action. Mitigate issues by performing operations directly on groups. We could issue run commands to our instances. We could run Patch Manager, patches. And we have a bunch of features in Systems Manager. No Parameter Store. Patch management. One I really want to point out, remote connect, key point without the need to open inbound ports, maintain Bastion hosts or manage SSH keys. And we're going to have a handful of lessons on Systems Manager, so we don't have to cover everything in this intro lesson. But let's just scout around some more. Get started with Systems Manager. So let's click Create. And there's a lot we can do in here. Notice interaction with AWS



Config. Let's just go with host management. So for host management, let's click Create. Manage your instances using Systems Manager. And we have some checkboxes here. Update Systems Manager Agent every 2 weeks. Collect inventory from instances every 30 minutes. Scan instances for missing daily patches. All good stuff. I think I'll leave that. And as you see, for our instances, we can install and configure the CloudWatch agent and make sure that it gets updated if we check these boxes. Likewise, update the EC2 launch agent. Then we have targets. So for targets, it defaults to entire organization. And yes, AWS Organizations is involved. We could customize that, only the OUs and regions we want, current account, and then we select Regions. I'm just going to go with this. We get a summary here of what we have enabled, and I'll just click Create. This takes a few minutes. All right, so it took about 2 minutes, and we have success. And I can click on this configuration and then view the details. We still have one pending, but we can still view details. So the collect inventory is not done yet. Sort of makes sense. It's dealing with multiple instances. Update Systems Manager Agent. Scan for patches. Enable Explorer. And there's that IAM role we want to attach. Now remember, we can create resource groups, and we can associate, say we're in a three-tier architecture, we can associate our instances with the back-end database and, again, group these resources and work with them through Systems Manager as a group. So there's a lot here in Systems Manager, and this is an introduction. We're going to have a handful of lessons. So I'll just leave the demo here. All right, let's talk about some of the controls and capabilities we have with Systems Manager. There are a bunch. I'm not going to read these word for word. But certainly, review these during your study time. So we can group our resources together for management purposes. Centrally define configuration options. Centrally view, investigate, and resolve. Automate or schedule maintenance tasks. Use runbook-style documents to perform actions on our instances. Run a command with rate and error controls. Again, we can connect to our instances without SSH or opening ports. Systems Manager Parameter Store. We can keep secrets in there. Inventory our instances. Metadata about our applications and network configurations. View consolidated inventory data and see which resources are out of compliance. There's that association with AWS Config. So there's a lot of capabilities with Systems Manager, and we'll try to build on this theme in the next five lessons. But that's the introductory look at Systems Manager. Let's take a look at some exam tips. What can we do with resource tags? Create resource groups and manage consolidated resources from a dashboard being Systems Manager dashboard. All right, some true/false for Systems Manager and its common distractors on the exam. AWS Organizations help you centrally manage and govern your environment as you grow and scale your resources. That sounds true to me. Trusted Advisor provides recommendations that help you follow resource group guidelines. That sounds a little bit iffy. The recommendations help you follow best practices. Systems Manager allows you to centralize operational data from one AWS service only. No way. It's certainly geared toward instances, EC2 instances or on-premises server. But as we've seen, other resources are



involved. Service Catalog allows IT administrators to create, manage, and distribute catalogs of approved products to end users. That sounds true. All right. And the point is, just let's have a good handle on these services and their use cases, and you often see them as distractors for Systems Manager. So with that said, we have a lot to cover on Systems Manager. Thanks for joining me, Cloud Gurus. I will see you in the next lesson.

Systems Manager Configuration Compliance

Hello, Cloud Gurus. In this lesson, we're going to talk about Systems Manager configuration compliance. Now the specific tool we're talking about is Systems Manager compliance. It's simply called compliance. Lesson title, I've added configuration. Our configuration in the AWS cloud we're trying to make uniform and compliant, and we can do that with the tools of Systems Manager. The tools of Systems Manager, they're all intermeshed and all work together. So it's difficult to choose which one to introduce first because, in this lesson, we'll talk about other tools in Systems Manager we haven't talked about yet, but we'll get to in upcoming lessons. So just bear with me on that. So in the lesson, we'll take an overview look at compliance and what it can do for us in our environment, and we'll talk about remediation. When we run into compliance issues, how we can quickly fix them and oftentimes in an automated fashion. And then we'll just look at some exam tips. So that said, let's look at Systems Manager compliance. All right, looking at our diagram, under the umbrella of Systems Manager, we have some new terms and tools. One of those, of course, is compliance, but also Patch Manager and State Manager. So let's define these first, and we will have an upcoming lesson on Patch Manager. So with compliance, it's a capability of Systems Manager, and we can scan our fleet of managed instances. And we're looking for patch compliance. So it integrates with Patch Manager and configuration and consistency. This integrates with State Manager. Let's quickly define these two first. State Manager is a capability of Systems Manager. Basically, what we can do with State Manager is to define a configure a standard configuration for our instances. Some things we can do with State Manager, bootstrap instances with specific software at startup, download and upload agents on a defined schedule, configure network settings, join instances to Microsoft Active Directory domain, run scripts on Linux, macOS, and Windows throughout their lifecycle. So we're getting our instances in a specific defined state, all dictated by State Manager. Compliance works with State Manager. Compliance displays current data about patching in Patch Manager and associations in State Manager. It's everything up to compliance. And it can go right to the source. It can scan our instances for patch compliances, and this can be done over multiple accounts and regions. It scans the instances for that information, and that information ultimately is displayed in Patch Manager and State Manager. Patch Manager is sort of self-explanatory, but let's close the loop anyway. Patch Manager is a capability of Systems Manager. It automates the



process of patching and managing nodes with both security-related and other types of updates. So we get our patches out to our EC2 instances using Patch Manager, and compliance can review our instances and make sure that they are up to date on their patches. Now if you're thinking, hey, this sort of sounds like work for AWS Config, you're absolutely right. And AWS Config can be in this loop as well. It gives us a history and change tracking for Patch Manager and State Manager data. Now we talked about Security Hub in a previous lesson. And Security Hub can come into play as well. Compliance data from Patch Manager can be sent to Security Hub. And we've all probably seen this at one time or another. We're using a popular application and there ends up being a security breach. How do they fix that quickly? They release a patch. So that's sort of where appropriate patching, up-to-date patching, comes into play. And it becomes a security issue, it can become a security issue, and can be reported in Security Hub. All right, so let's say we find some issues. How do we remediate these issues? And the theme of the course, automation. Can we automate remediation of our issues? And you probably know that the answer is yes. Issue remediation comes into play. Compliance is a part of State Manager. Compliance issues reported back to Systems Manager can trigger remediation tasks. And there's that magic word, triggers. How do we trigger SNS? Well, certainly, some of our findings in compliance can ultimately trigger email notifications. We can trigger Lambda functions. And with that, obviously, we can put notes into Slack. All of our common trigger mechanisms can be used. So that's our notification, but can we actually fix issues? Absolutely, we can. EventBridge. Obviously, EventBridge events can be triggered. We can have scripts. Let's say we have a compliance issue triggers EventBridge. EventBridge could kick off scripts that would run on our instances and bring them into compliance. Additionally, Systems Manager Run Command, which we'll talk about in the next lesson. It's really sort of the same process, different vehicle. We can execute Run Command to get our instances into compliance. So again, we can configure EventBridge to performing an action in response to Systems Manager compliant events. We can quickly remediate patch and association compliance issues by using the Run Command. Again, we'll deal with Run Command in the next lesson. So just trust that we can do it. And then in the next lesson, we'll get into the mechanics of how Run Command works. Much we can do with Systems Manager, and it looks a little bit intimidating at first, but it's really very straightforward. And there are interactions, but they're all straightforward. And if we look at them one at a time, break it down step by step, it's not so intimidating. So in this lesson, I wanted to talk about compliance and what we can do with compliance. It's looking for our patch compliance. It's working with Patch Manager and also Security Hub. Make sure that we're up to date on our patches. We can get up to date if we need to be with our automated events. And remember, it can also work with State Manager to ensure our instances are in line with the state that we define for our instances. They have all these software updates. They have all of their patches. Compliance can check that for us, and we can remediate if necessary. So let's look at some exam tips. Again, we're



checking for patch and configuration inconsistencies with Systems Manager compliance. How do we remediate issues that we find? Our inconsistencies, our findings, can trigger Run Command and/or EventBridge, and we can get up to date with State Manager by updating our instances. And remember that other connection at the pro level. We're always looking at interactions between services, not just what the service itself can do. Compliance findings can be sent to Security Hub for a comprehensive view of compliance status. I'll emphasize again, often patches are to fix security breaches. So if our patches aren't up to date, it definitely can be a security issue. So that's a look at Systems Manager compliance. And if you understand the interactions it has within Systems Manager and with other services presented in this lesson, you will be just fine for the exam. So with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Systems Manager Run Command

Hey, Cloud Gurus. Welcome. We're going to talk about Systems Manager Run Command. And with Run Command, we can remotely manage configuration of our servers, even if those servers are on-premises. So we can have a hybrid environment on-premises servers and EC2 instances in our cloud environment and manage them all the same. And for the on-premises servers, I'll use an analogy. If we want on-premises servers to work with CloudWatch, we install the CloudWatch agent. Well, if we want those on-prem servers to work with Systems Manager, we install the Systems Manager agent on those on-premises servers virtual machines, and we're ready to go and work with Systems Manager. So in this lesson, we will take an overview look at Systems Manager, and then we're going to do a demo to show the capabilities of Systems Manager. And we will wrap up with some exam tips. Should be a short but useful lesson, and we're starting to tie all these Systems Manager pieces together. So let's talk about Run Command. All right, how can we work with Systems Manager? Well, programmatically, using SDKs, from the management console and from the CLI. I debated what to use for the demo, and I chose to use the management console. I just like the bells and whistles that it gives for a demo. A lot of good visuals to help our retention. And also, we'll have a couple Systems Manager hands-on labs, and I didn't want to trample all over those labs. So we'll just work from the management console. But all the same concepts apply whether we're accessing Systems Manager programmatically through the management console or through the CLI. Now as far as Run Command, as you've come to expect, there are some IAM privileges that we would need to configure. And what we need to configure is an instance profile. So with the instance profile, we are attaching an IAM role to our EC2 instances at launch time. And within that role, of course, are IAM policies. The main policy, AmazonSSMManagedInstanceCore, pretty much allows an instance to use Systems Manager core functionality. But of course, we can customize policies and get those policies to meet our needs. And from



Systems Manager, we can issue the Run Command. This is showing Systems Manager documents. Now we can use prebuilt documents that we'll see in the demo. We'll see in the management console these prebuilt documents. Or we can create our own custom documents, and these documents are called command documents. And again, we will be using one of these in the demo. So with Run Command, we can issue commands to our instances. I'm showing AWS in a VPC. Remember, we can do this with on-premises servers or VMs. We simply need to install the Systems Manager agent. Now the thing we can do is issue commands to our containers. And just to reiterate some key points, Run Command lets us remotely and securely manage the configuration of our managed instances. Remember the lesson on compliance. If we are out of compliance, we can automate a Run Command to get our servers back into compliance. So we can install our Bootstrap applications, build deployment pipelines, capture log files when an instance is terminated, and join instances to a Windows domain. Those are some of the things we can do. So that's a straightforward look at what we can do with Systems Manager Run Command, and let's just fortify that understanding with a demo from the management console. All right out at the management console in Systems Manager. But, first, let's start at IAM. And we want to create a role for Systems Manager. So I'll click on Roles. Create a role. AWS service. And we actually want EC2. All right, EC2. There is a role for Systems Manager, but I'm going to stick with this one. Click Next. I want AmazonSSMFullAccess for the policy. Click Next. Give our role a name and create. All right, so our role has been created. Next, let's go over to EC2 and create an instance, Amazon Linux, free tier eligible, Linux 2023 AML. Instance type, going with t3.micro. Key pair, I'm going to proceed without a key pair. All right, under Advanced details, we want to assign that role we just created. There it is. IAM instance profile, this role we just created. All right, let's go back up to the top, name and tags. I'll give it a tag of Name, value ssmDemo, and let's launch this instance. All right, now at Systems Manager in the management console. Now we just launched an instance, but let's say we did that a month ago. And we want to update the SSM Agent on any instances that need it. Well, we just have one instance, but that'll work. So let's scroll down till we get Run Command. Let's select that. And we want to run a command. I searched on AWS Update. We want this UpdateSSMAgent. Now we could specify a version. But if we leave it blank, it's going to update with the latest version. We could downgrade to an earlier version. All right, the important part, target selection. Let's bounce over to EC2 real quick. I've selected our instance that we just created, selected Tags. Here's our key and our value. Back over to Systems Manager. So this identifies the instances. Again, we only have one. Multiple instances could have this tag. We have comments, optional. There's a timeout, rate control. We only have one target. Write command output to an S3 bucket. And I'll just choose bucket for this course. We add CloudWatch alarms, SNS notifications, but I'm just going to run it. All right, in progress with one target, basically our EC2 instances. All right. So our update is in progress. Not even going to hang around for that to complete. I have faith in that. So remember, command



description here. We're updating the SSM Agent. Key points are we need to create a role. Attach that role to our EC2 instance. And yes, we only updated one instance. But with tagging, we could be applying this update to many instances at one time. All right, let's look at some exam tips. What is Run Command? Secure remote management of our instances at scale without needing to log in to our servers. We don't need bastion hosts or SSH or remote PowerShell. Common administrative tasks. Registry edits, user management, software and patch installations. We installed Apache Server in our example. How can we control who can execute commands in Run Command? IAM policies. Remember that instance profile as well. Policy is inside of the role. The role is in the instance profile, and the instance profile can be attached to our EC2 instances. So that is Systems Manager Run Command. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Systems Manager Parameter Store

Hey, Cloud Gurus. Welcome. In this lesson, we're going to take a look at Systems Manager Parameter Store. Parameter Store is another capability of Systems Manager that provides secure hierarchical storage for configuration data management and secrets management. Think of passwords, database strings, AMI IDs, license code, access keys, and secret access keys. And back in the day, these secrets would be stored in our code, and that was quickly hacked and deemed to be a bad idea. Now in fairness, developers knew that and often stored these in configuration files, which the code can then access. But with Parameter Store, AWS provides us with this vehicle to securely store sensitive information. So in this lesson, we'll identify the problem. We've already talked about it a bit. Take an overview look at Parameter Store. Go and do a demo to solidify our concepts, and then we'll talk about some exam tips. So let's go ahead and identify the problem. And this diagram really simplifies it, I think. So, we have a pipeline here, CodeBuild and CodeDeploy. We're using a build spec file. And guess what? Within that build spec file, we're storing an access key, a secret access key, and a database password. Is it a good idea to store sensitive data files in our build spec file? Well, we know the answer to that. Not a good idea at all. So along comes Parameter Store. And basically, we can store our sensitive information in Parameter Store and then reference this information from our code. In a nutshell, that's what Parameter Store can do for us. Now before going on to the demo, let's just take a different look here, an overview look. All right, so we have a situation where we want to retrieve a database connection string. Rather than store that in our code, we do it the right way. We store it in Parameter Store. And then our instance can retrieve that from Parameter Store and use that to access our RDS database. So again, Parameter Store provides secure hierarchical storage for configuration data management and secrets management. And we can store such things as passwords, database connection strings, instance IDs, machine image IDs, license code. And think of



access keys and secret access keys, we can store this all as parameter values. And then we can reference these parameters in our scripts, our commands, our documents and configuration and automation workflows. Now, in the demo, we're going to reference a parameter from a CloudFormation template. Infrastructure as Code, it's not a good idea to store that information in our templates. So let's tie these concepts together with a demo. All right, so I'm in the management console in CloudFormation. I also have a Systems Manager tab open. This template creates an EC2 instance and a security group. Now notice in the template that we have some parameters that we're going to enter up front during stack creation, a key pair instance type. And the instance type is hardcoded, and we have a lot of options. An additional parameter is our SSH location, but that's not important in this demo. What is important is this instance type. So, to demonstrate what we can do with Parameter Store, we're going to store the instance type in Systems Manager. So let's go ahead and do that. Let's create a parameter. I'll call it devinstance. All right, so we can select tiers. We're just going to keep default. We want a string value, and let's go t2.micro. Go ahead and create the parameter. Let's create another parameter. We'll call it prodinstance, and we'll go t2.small. And the thinking being so we select whether we want a dev or prod, the prod instance is more powerful. Now showing how we can store these instance types in Parameter Store is certainly useful if we want to build a dev or prod environment. But this could just as easily be database passwords that we're extracting from our template and putting in Parameter Store. I couldn't bring myself to do that. It's just really bad practice to have passwords in a template. All right, so we have these parameters created. Let's go back to CloudFormation and into our template. And what I'm going to do is zap everything. Delete everything under Instance type. Now we still need an instance type. So our instance type is going to be `AWS::SSM::Parameter::Value` string. So what we are entering in the parameter when we create the stack is we're either going to enter that dev instance or the prod instance. Then I enter a default value, devinstance. The reason I like demoing this instance type is, well, if this blows up, we know where the problem is. And it's very binary. We either get an instance, or we don't get an instance. Another cool thing about pulling instance type out of our template and into Parameter Store, let's say we're using this parameter in, say, 20 templates. But let's say that we underestimated our traffic. We need bigger servers. What do we do? We come over to Parameter Store, and we make the change in one place. We could bump up this value, and it propagates throughout all of our templates. Let's see this in action. Let's create our stack, validate my template, and let's run this. Everything's good. Let's click Next. So here's our new type of parameter drawn from Parameter Store. I'll change that to prodinstance. Here's our old school parameter, our key pair SSH location. Always good to name our stack. Now we can run it. Keep the defaults. Let's go ahead and create our stack. And remember, we need an instance type. We're drawing the instance type from Parameter Store. If something goes wrong, we don't get an instance. And of course, we know the lines of code to look at in our template. All right, it's created the security



group, and then it will create the instance, or it will fail. All right, create complete. We have our instance and our security group. Again, I wanted to demonstrate using the instance type. Very useful. We could create a dev or prod instance based on the parameter entered and also just showing that code reuse. If we want to bump up the servers, beef them up so to speak, we change one value in Parameter Store. We could be using this parameter in 100 templates. We make the change in one place. Now, of course, we could just as easily do this with, say, database passwords or connection strings, and that's how we would pull those out of our templates. All right, let's close out the lesson with a couple exam tips. Know the purpose of Parameter Store. Purpose is to provide a centralized store to manage your config data, whether it's plaintext data, database strings, or secrets, such as passwords. And we saw a demo of using instance type in Parameter Store. Can we encrypt sensitive data in Parameter Store? If so, how? Yes. We can use custom KMS encryption keys or AWS KMS encryption keys, and we can encrypt our secrets in Parameter Store. And with that, I'm going to end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Automate with Systems Manager - Demo

Hey, Cloud Gurus. Welcome. In this lesson, we're going to talk about automation with Systems Manager. Now in the entire course, our theme has been automation. But in Systems Manager, automation is a thing. Automation is a capability of Systems Manager, just like Run Command and compliance. And automation can simplify common tasks, maintenance tasks, deployment and remediation tasks for EC2. But not only EC2, other services such as RDS, Redshift, S3, and many other services. So we'll check out automation in this lesson. We'll start with an overview look at automation, and then we'll check out a use case just to ingrain these concepts. And to further ingrain the concepts, we'll do a quick demo, and then we'll wrap up with some exam tips. So with that, let's go ahead and check out Systems Manager automation. All right, so we have a mostly built-out diagram here. We have Systems Manager, Automation. And automation runs automation documents. And recently, they have been renamed to runbooks. So as an example, let's say we have some EC2 instances in our environment. And we're using these EC2 instances for some specialized batch processes. Now, when they complete the batch, we want to destroy these instances. And maybe they're running over the weekend. Nobody's working, but we can set up an automation. So, we can have EventBridge monitoring these instances. And EventBridge is triggered when our application finishes. EventBridge can then trigger a Systems Manager automation. And in this case, the automation would be to terminate these instances. So that's the type of thing we can do with automation with no manual user intervention. Of course, we have to set up the automation, which we'll take a look at in the demo. So with automation, we can perform common IT tasks such as stop EC2



instances, again, terminate EC2 instances. And just for a little something different in the demo, we will restart EC2 instances. Other tasks, perform disruptive tasks in bulk. So we could update CloudFormation stacks all at once. Again, restart EC2 instances. Update AMIs. And just realize that, yeah, we can go into the management console and run an automation, which we'll do in the demo. But they can also be triggered and run automatically. So let's check out a use case for automation. All right, problem. We need to curtail spending on our servers and databases. We've considered shutting down our servers during weekends and off hours. But due to some requirements, shutting down instances for long periods of time is not an acceptable option. Shutting down the database on the weekend is acceptable. Can we downsize? If so, how can we automate this? So we can schedule an automation, say Friday evening, that can downsize EC2 instances and stop the databases. Now, that's half the battle. Late Sunday night into Monday morning maybe, we also need to schedule an automation to scale back up sort of to the working week size of our instances. So we can automate this with Systems Manager automation. We can schedule these automations. And it's just like setting a timer on an oven. All of these capabilities in Systems Manager are pretty straightforward and actually really interesting and fun to work with. Let's look at a demo from the management console on Systems Manager automation. All right, so out at Systems Manager. And if we look on the left here, we'll see Automation. And it's showing past automations that I've done. Let's do a new automation. Execute automation. And we can narrow things down here by checking some boxes, but we can also search. And this is the automation we want, RestartEC2Instance. Now I opened another tab over EC2. I have two instances running. So let's click into this one. And notice, it has two actions, stopInstances and then start. And that's what we want. And here is contents of this runbook. Not surprisingly, JSON, mainSteps, stopInstances, startInstances. Details, your EC2 instance, ID. Optional, ARN of the role allows automation to perform the actions on your behalf. All right, so let's execute this automation, and we have some options here. Single execution, execute on targets. Now if we had a bigger operation, more instances, we could do rate control. Execute safely on multiple targets by defining concurrency and error thresholds. I like to analogize that's a lot like CodeDeploy. We could do multi account and region. Just to see, I'll check that notice. No surprise here. Organizations is involved, regions as well. Manual execution. Let's just go with that one actually. So I have two instances. And just to draw a contrast, I will just select one so that only that one gets stopped and restarted. We could set up an alarm. Add tags. Let's go ahead and execute this automation. Overall status, waiting. Still waiting. What's going on here? Well, we're doing a manual execution. So, we're at step 1. Let's go ahead and execute. All right, in progress. I'll switch over to EC2. Probably won't take long. Ah, let me take off this filter. Here's our instance, stopping. I'll just bounce back and forth, in progress. All right, so our instance is stopped. That took about a minute. Let's go back. All right. So we're back in a waiting state. Step details. Success. So basically, our EC2 instance is stopped and we verified that. Now remember this is manual. So we need to



execute step 2, which is restarting that instance. Let's go back over EC2. Click Refresh. It's running. So are we done? Let's check, in progress. Well it says its running, right? Why would we not be done? It's waiting on these status checks. Back over to EC2, and we'll just wait on this. All right, the status checks are completed. And sure enough, step 2 is done. Our instance is restarted. Lets click Create. Review results. Looks like that took about a little over 6 minutes. And there are two steps. So this could be a progression. We did this one manually. We did it on one instance. And we could do it automatically and finally roll it out to a larger group of instances, even in multiple accounts and multiple regions. So that's how we work with automations in Systems Manager. So let's wrap up our lesson with a few exam tips. Couple brain teasers here. Automation documents are now called runbooks. They define the actions the automation performs on EC2 instances or other resources. In the case of the demo, step 1 was to stop the instances. Step 2 was to start them. Automations can work with EventBridge. EventBridge can trigger our automations. What is a target for EventBridge? Systems Manager can be a target for EventBridge. Automation actions will trigger EventBridge, which can then update Systems Manager. So EventBridge can be both an input trigger to Systems Manager and an output. So an automation action could actually trigger an event in EventBridge. Now, we can circle back and update something in Systems Manager or maybe can trigger a Lambda function. So that is a look at Systems Manager automations. Automation is not just our theme for the course, it is also a capability in Systems Manager. And with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

AWS Systems Manager Session Manager

Hey, Cloud Gurus. Welcome to this lesson, AWS Systems Manager Session Manager. So hopefully this is a review on Session Manager. But if not, we have you covered with this. And we'll cover some real key points on Session Manager. So what is Session Manager? It's a fully managed AWS Systems Manager capability where you can manage your EC2 instances, edge devices, on-prem servers and virtual machines using browser-based shell or the AWS CLI. All right, use case. Wow, we've just gotten a strict lockdown from corporate on accessing EC2 instances. No open ports, no bastion hosts, no SSH key, no problem. We can use Session Manager to access our EC2 instances, edge devices, on-prem servers, and virtual machines. And it looks a little something like this. What do we need? We need the Systems Manager agent. Certainly, would install that on your on-prem server. As we'll see, some of our AMIs from which we create our instances already have that agent. All right, what are the key features of Session Manager? Well, it works with Windows, Linux, and Mac. We can access our instances from the CLI and SDK using IAM access control. We can do logging and auditing. We'll show how to set that up in the demo. Shell profiles, customer encryption, and it has AWS PrivateLink support, tunneling, and interactive



commands. All right, so how can we manage EC2 instances from Session Manager to key components. An EC2 role for SSM. And when we launch an EC2 instance, we'll select an instance profile attaching this role to our instance. And we need that Systems Manager Agent, otherwise known as the SSM Agent. And with those two key components, we can manage an EC2 instance. And in the demo, we'll set this up. So let's do that. All right. Out at EC2 in the management console, I'm going to launch a new instance. Give that instance a name, Amazon Linux 2023. And I'm going to go with a t3.micro. Not going to use a key pair. Go down here to Advanced details and select an instance profile. All right, now there's an instance profile, a role that I created. And this role has an SSMFullAccess policy. Now there's another policy for managed instances. The full access also has CloudWatch capabilities, and I want to show setting that up as well. And we can just accept the defaults otherwise and launch our instance. View instances. All right. It's initializing. So I've opened another tab, Systems Manager. And down below here, Session Manager. Click that. And we want to configure preferences. Note the idle session timeout, 20 minutes. Personally, that's a bit much. I can live with 10 minutes. All right, notice we have some logging here, CloudWatch and S3. We're going to deal with CloudWatch. So when we enable that, it expands. Now I'm going to go over to CloudWatch and going to Log groups and create a log group. Okay, just give the log group a name and click Create. All right, back over at Session Manager. We want stream session logs, and I can just page through or search and find the log group we just created. All right, all set. We can click Save. Back over to Sessions and click Start session. So I select my instance and click Start session again. We're connected to our instance. Just did an ls command? Got a heck of a lot of files, pwd. I'm in user bin. Just executing a few commands, host name command. All right, that's going to be enough activity to serve our purposes. Going to terminate. Back in Sessions, let's go to Session history. And yes, as they say in the movies, that just happened. Now, it's a little bit sneaky. It's kind of hidden, but notice we can scroll, and that's important cause, from here, we can bounce over to CloudWatch Logs. All right, so we have some info here, but we can expand. There's my ls command. There's my pwd command, host name command. All right. So in that way, we are able to use Session Manager to manage our instances. He points the instance profile and the SSM Agent on our instance. So what about that agent? Well, let's go back over to Sessions. Let's start another session. Same instance, of course. Obviously, it worked. We have the SSM Agent. But if you're in doubt ever that your instance has the agent, this is your command, sudo systemctl status amazon-ssm-agent. So in green, enabled, and it's active and running. So Amazon Linux 2, Amazon Linux 2023, you're going to have this agent. But again, that command can check for you. So that's a look at managing instances using Systems Manager Session Manager. With that, I will end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

AWS Systems Manager - Default Host Management



Configuration

Hey, Cloud Gurus. Welcome to this lesson, AWS Systems Manager - Default Host Management Configuration. In the last lesson, we set up an instance to be managed by Session Manager and key points were proper instance profile and the SSM Agent on the instance. But we can set it up so that our instances within a region are automatically managed by Systems Manager. Let's see how this works. All right, what is default host management configuration? This feature allows Systems Manager to manage your instances automatically as managed instances. Once we have this set up, we don't have to do anything else for those instances to be managed by Systems Manager. Now what do we mean by a managed instance? A managed instance is an EC2 instance that is configured for use with Systems Manager. And certainly, right off the bat, we know it has to have that SSM Agent on it. If it doesn't, we would have to install it. But we can configure our environment so that instances created in our region automatically will be hosted by Systems Manager. What are the benefits? Well, referring back to the last lesson, we can connect to our instances securely using Session Manager. And yes, we can set that up manually. But with DHMC, it's set up automatically for us. Perform automated patch scans using Patch Manager. View detailed information about your instances using Systems Manager Inventory. And track and manage instances using Fleet Manager. All right. So how do we set up DHMC? There are a couple of key points. An instance to be managed must use Instance metadata Service version 2. There's also a version 1, and we'll see how we can find this out. It's actually pretty easy. And then the SSM Agent must be installed on the instance to be managed. So again, Amazon Linux 2, Amazon Linux 2023. We checked that for 2023 in the last lesson. We're going to be working with that same instance. It already has the agent. Now here's a key point. If an instance profile is already attached to an EC2 instance that is to be managed using Systems Manager, you must remove any permissions from it that allow the SSM update instance information operation. If you allow SSM update instance information in your IAM instance profile, the instance will not use the default host management configuration permissions. So basically, we just have to be careful. And we can set up an instance profile for this. I'm not going to do that in this demo. But in the policy attached to our instance profile, we would just have to make sure that this update instance information operation is not in policy. So let's see how we set this up. All right, I'm going to start at EC2. And I'll go to Instances. I don't have any instances running, but I have several in a stopped state and that's okay. All right, so what we need to do is come over here, top right, and click on our Settings icon. So we have this Attributes column. And it might not jump out at you, but that is scrollable. And this setting, IMDSv2, we want that on. There's an IMDSv1. We don't want to use that anymore. Then we can click Confirm. And again, this panel is scrollable also to the right. Doesn't jump out at you, but it is. So, IMDSv2, when it says required, well, we have IMDSv2. One of those is optional and required means that when you request metadata from



this instance, IMDSv2 will work. v1 will not work. Optional means that either v1 or v2 will work. So again, to set up default host management, we need v2. All right, let's go over to Systems Manager and see how we can enable DHMC. So let's click on Quick Setup. And we want to go to Library, and we've seen this page before. And if we just scroll down, there's our default host management configuration. Enables the default host management configuration for all accounts and regions in your organization. Let's click Create. For turning on this setting, ensure the following requirements are met. Latest version of SSM Agent is installed on your instances, and your instances are using IMDSv2, and all of them are. The one is using both v1 and v2, the one that said optional. All right, additionally, configuration options, enable automatic updates of the SSM Agent every 2 weeks. So, we're setting up this automatic management, and we can automatically update the SSM Agent. Summary information, just note the involvement of AWS Organizations. And let's click Create. All right and this just took a couple of minutes, but we have success. But we have successfully set up default host management configuration. And with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

AWS Systems Manager Hybrid Environments

Hey, Cloud Gurus. Welcome to this lesson, AWS Systems Manager - Hybrid Environments. And by hybrid environments, we're talking about managing both EC2 instances and on-prem servers. And maybe a better description of that would be EC2 instances and any number of non EC2 machine types. So in the lesson, we'll talk about how this is set up and do enough of a demo so that we understand some terminology. So let's see how we set up hybrid environments for Systems Manager. All right, what's a hybrid environment look like? Systems Manager can manage both EC2 instances and a number of non EC2 machine types. And think on-premises servers. Any machine that has been configured for use with Systems Manager is called a managed node. We'll see there's a difference in the notation between EC2 and non EC2. So of course, we can manage our instances in our AWS regions, both Linux and Windows instances and, of course, multiple regions. But additionally, let's say we have an on-premises data center, we can manage those servers as well through Systems Manager. Let's talk about setting this up, and we'll have a quick demo. Just really want to show a couple of screens here. We create an IAM service role for a hybrid and multi cloud environment. That's part of the demo. Just want to see what that looks like. And then we create a hybrid activation for a hybrid and multi cloud environment. Really want to see what that looks like. That's part of the demo. As you would expect, we would install the agent on our on-prem servers. And the above step can be done for both Linux and Windows environments. So let's jump out to the management console and just take a look at these first two steps just to make sure that makes sense. All right, out at the management console in IAM. Let's go to Roles and Create a role. Trusted entity type, we'll keep AWS



service. Use cases we want Systems Manager. And we'll keep that radio button where it is, Systems Manager. Then we can click Next. And I'm just going to search on SSM, and we want AmazonSSMManagedInstanceCore. Okay we've seen this policy before. This is for, among other things, managing EC2 instances. But let's say we also want to manage on-prem machines, and we're using Active Directory on-prem. We want this policy also, AmazonSSMDirectoryServiceAccess. And let's just click on that. This policy allows SSM Agent. Of course, that has to be installed on that server to access directory service on behalf of the customer for domain-join the managed instance. Now we could add some other things, EventBridge or CloudWatch. But, most importantly, with these two policies, we're managing EC2 and we're managing on-prem servers that are using Active Directory. So let's click Next and just give our role a name. I'll just call it SSMHybrid and create the role. So that is step 1. We create that hybrid role. Just wanted to show that just to make sure there's an understanding of what's going on and why. Now what about the hybrid activation? Step 2 is create a hybrid activation for a hybrid and multi cloud environment. To set up these non EC2 machines, these on-prem machines, as managed nodes for hybrid environment, we have to create this hybrid activation. And when we create this activation, we immediately receive an activation code and an activation ID. The next step beyond that, which I'm not going to show, would be to install the SSM Agent on non EC2 machines. I don't have my own data center, nor do I have any non EC2 servers right now. But if I did, install the SSM Agent. And we would specify when we do that, an activation code and activation id, which we will receive. So let's see what that looks like. Let's go over to Systems Manager. And down here, Hybrid Activations. And let's create an activation. I'll just enter an optional description. Specify the number of servers we want to register. And I'm going to keep this radio button here. If you select this option, AWS creates a new role for you named AmazonEC2RunCommandRoleForManagedInstances. For activation expiry date, must be in the future and not more than 30 days in the future. Then I'll just add a name here and click Create activation. All right. So we had success. And notice, your activation code is listed below. Copy this code and keep it in a safe place as you will not be able to access it again. So we would do that. And then for our on-prem server, install the agent, and we'll be able to add this activation code and activation ID. Not going to show that step. I don't have any spare servers laying around. But those are the key steps, the service role and the hybrid activation. All right, so once all of that is done, those servers will show up in Systems Manager, and we can manage them just as we would EC2 instances. I just want to show a few more things to wrap up. So we have our managed nodes. Here's a key point. Hybrid activation machines are listed in the management console and described as managed nodes. Here's the difference. EC2 instances, and we've certainly seen this, are prefixed in the management console with an i dash and then that ID number. Hybrid machines are prefixed in the management console with m dash and then an ID number. That's the difference. Otherwise, they are nodes in our account that we manage just like EC2



instances from Systems Manager. And with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

AWS Systems Manager Advanced

Hey Cloud Gurus. Welcome to this lesson, AWS Systems Manager Advanced. And don't be misled by that word advanced. Not anything harder than what we've seen already. It's just that these topics aren't the first things you would do in Systems Manager. So let's talk about some of these lesser-used features and certainly features that you may see on the exam but not heavily featured. Let's first talk about Systems Manager with AWS IoT Greengrass, which begs the question, what is IoT Greengrass? IoT Greengrass is an edge runtime and cloud service that helps you build, deploy, and manage IoT applications on your devices. And maybe the next question you're asking, I'll answer. Yes, you can manage these devices through Systems Manager. Greengrass core devices support all Systems Manager capabilities. And by this time, you'll probably know the answer to this question. How do we manage these devices through Systems Manager, the Systems Manager Agent. The agent can be installed so you can manage core devices of IoT Greengrass through Systems Manager. And that's really as deep as we need to take it for the exam. So let's move on to the next topic. SSM OpsCenter. OpsCenter is a capability of Systems Manager that provides a central location where operations engineers and IT professionals can manage operational work items, otherwise known as ops items. An ops item is any operational issue or interruption that needs investigation and remediation. Contextual investigation data about each ops item can be viewed using OpsCenter. You can also use Systems Manager automation runbooks to resolve ops item. These ops items include relevant information such as name and ID of the resource that generated the ops item. Now we can integrate OpsCenter with other AWS services, and it will create these ops items automatically. Those services include AWS Config and EventBridge. So in addition to OpsCenter, we can use all of these services to investigate an ops item. The key point is this can all be done in OpsCenter. Once these other services are integrated, the information is going to end up in OpsCenter. Can we use OpsCenter with our on-prem managed nodes? Yes, absolutely. So again, once Systems Manager is integrated with those managed nodes, ops items for those managed nodes will appear in OpsCenter. All right, one more, Systems Manager with VPC endpoints. So we're accessing our servers, EC2 and our on-prem servers, through Systems Manager. Can we tighten up our security for this interaction? And the answer is yes, absolutely. And we can do that with VPC endpoints. We certainly can integrate these endpoints with Systems Manager. And by using a VPC endpoint, we can connect to services powered by AWS PrivateLink. PrivateLink allows us to privately access our EC2 instances and Systems Manager APIs by using private IP addresses. So basically, our managed instances don't have to traverse the public internet. So if we have PrivateLink, we don't need an internet



gateway or a NAT device or a virtual private gateway. So Systems Manager will go through this VPC endpoint to our managed instances. What's missing? I'll bet that's not a shock. The Systems Manager agent needs to be installed on EC2 instances. And we've learned with Amazon Linux 2023, the agent is on there. So that's a look at some advanced topics. Not really extremely difficult, but probably not the first way you would use Systems Manager when you start using it. You may see these mentioned on the exam but not heavily. Just understand how they work and what their purpose is, and you'll be fine. So with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Security and Compliance Services

Trusted Advisor for Policies and Standards

Hey, Cloud Gurus Welcome. So in this lesson, we're going to take a look at Trusted Advisor for policies and standards. And this domain covers a lot of sort of complementary services that we're aware of, but we don't use too often. And Trusted Advisor certainly fits into that. We've probably seen Trusted Advisor by this point, but it's good to review because you will see Trusted Advisor in scenarios, both as part of a correct solution but also often as a distractor. So in the lesson, we'll take an overview look at Trusted Advisor and talk about the support plans, different levels of support we can get for Trusted Advisor. And we'll do a quick demo just to check it out from management console and ingrain those concepts. Take a look at a use case just to tie it into the DevOps philosophy and how it fits in there and how we automate. And we'll end with some exam tips. So let's check out Trusted Advisor. So Trusted Advisor is always analyzing our environment and providing feedback for us and some best practices. And what areas does Trusted Advisor provide that feedback Cost optimization, performance, security, fault tolerance, and service limits. Does it also provide recommendations? Yes, absolutely. It provides recommendations and action links. So it may provide you with, for example, a recommendation that a security group is too open. And it would provide a link where you could go and rectify that situation. Now depending on the support plan you have, there's various levels of recommendations you could get. Let's talk about the support plans. There's the basic support plan, developer, business, and enterprise. And the business and enterprise support plans provide a more robust version of Trusted Advisor. Let's check that out real quick. So basic and developer are going to provide us the seven core Trusted Advisor checks; whereas business and enterprise provide the full set of Trusted Advisor checks. And this link here, you can go out to AWS and check out the full list of those checks. If you're an individual, you probably have the basic plan. But let's check out the seven core checks. Trusted Advisor provides these. Even in the basic and developer plans, it'll check your bucket permissions, provide recommendations, security groups. So if you have a security



group and the good old 0.0.0.0 open to the whole world, Trusted Advisor will point that out to you. We'll report back on IAM use. Do you have MFA on your root account? Do you have public EBS snapshots and also RDS public snapshots and also service limits? So that's you get out of the box with Trusted Advisor without even upping your subscription. Again, this is sort of a review lesson. We should know about Trusted Advisor, but it turns up on the DevOps Pro exam a fair amount. Again, sometimes in the proper solution and sometimes in the incorrect solution as a distractor. We just want to review and know what it can do for us. And with that said, let's check out what it can do for us from the management console. All right, so I'm a Trusted Advisor in the management console in the dashboard. And note, my major checks here, cost optimization, performance, security, fault tolerance, service limits. And I have a couple of findings. I don't have MFA on my root account. And this one we talked about. I have a security group with a port which is open to the whole world. Action recommended on these two and then did a check on my bucket permissions. So remember, Macie checks S3 buckets, but it's checking the data, and Trusted Advisor is simply doing a check on some bucket permissions. And we can drill down into each of these. Notice that upgrade our support plan, we get a lot more checks, these grayed out areas. And it seems like security is the one that is the most useful for folks with the basic plan. Fault tolerance and service limits. Service limits is pretty darn useful as well. So just familiarize yourself or refamiliarize yourself with Trusted Advisor. Again, it turns up fairly often on the exam as part of a scenario question. One other thing I want to show. So in EventBridge, create rule is Trusted Advisor. And event source, as you might expect, the answer is yes. So again, it can be part of our automations with EventBridge. So that's a look at Trusted Advisor from the management console and also just a refresher again on EventBridge. All right, so we saw how we can link Trusted Advisor with EventBridge. We can also link it with CloudWatch Alarms. Can we be automatically notified when we have a service limit issue for EC2 instances in our account? Yes. There's a Trusted Advisor metric in CloudWatch Alarms, service limit usage. So just understand the difference. In Trusted Advisor, we can manually view the service limits. But we can only introduce automation with our CloudWatch alarm. So if we went into Trusted Advisor, we could see the service limits, but this will notify us automatically. Can link that up with SNS and be notified. And some of the same suspects here. We could kick off a Lambda function. But this is the sort of use case where Trusted Advisor would come into play in a scenario question. All right, let's take a look at some exam tips. Key terms for Trusted Advisor, best practices, recommendations. So, if you see these terms in a scenario question, you want to start considering Trusted Advisor as part of the correct answer. All right, regarding scenario questions, what are some common distractors for Trusted Advisor? AWS Inspector, Cost Explorer, GuardDuty, and CloudWatch. Although, as we see, they can work hand in hand, but they have to be linked together for the automation to take place. As always, know these simple use cases. It will really help you out in scenario questions in getting the correct answer. But even more so, remember, there's



going to be three or four incorrect answers. And what categories does Trusted Advisor check? Cost optimization, performance, security, fault tolerance, and service limits. So that is a look at AWS Trusted Advisor. And with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

Patch Manager

Hey, Cloud Gurus Welcome. So in this lesson, we're going to talk about Patch Manager, and Patch Manager is a part of Systems Manager. And with Patch Manager, we can automate the process of patching managed nodes with security-related and other types of updates. And recall from the compliance lesson, we would be able to send compliance information on Patch Manager to Security Hub. So again, patching is definitely a big part of our tightening our security. Take an overview look at Patch Manager. And then we're going to go ahead and do a demo, a little bit different demo. I have a couple demos in this section where we're performing operations on our EC2 instances. In this demo, we're going to do something a little bit different, sort of a preparation of what we're going to do on our EC2 instances. Just a little bit different look. And then finally, we'll take a look at some exam tips. So let's take an overview look at Patch Manager. All right. So under the umbrella of Systems Manager, a capability of Systems Manager is maintenance window. And with maintenance windows, we can help define our schedule for when we want to perform potentially disruptive actions on our nodes, things like patching an operating system or updating drivers, installing software, and, of course, in the context of this lesson, patches. And maintenance window works with other services such as S3 buckets, SQS, Key Management Service. So not just EC2 instances. Now, for patching naturally, we're going to be patching on our instances. And remember, all we have to do is install the Systems Manager agent on on-premises servers to operate on those servers with Systems Manager. When we add the agent, there's always some IAM setup that needs to be done. Now recall in the Systems Manager Run Command lesson, we talked about control documents. One of those control documents is RunPatchBaseline. So if we went into Run Command, like in the previous lesson, this document runs. It's going to use the patch baseline specified as the default for the specific operating system unless we tell it to do otherwise. And this document supports Linux, macOS, Windows Server. Again, RunPatchBaseline can be automated to apply patches to the appropriate instances. Now, I'm showing Systems Manager here, just emphasizing that Patch Manager is a capability of Systems Manager. Remember how all of these capabilities in Systems Manager worked together. Remember the compliance lesson. Compliance can report if we're out of compliance. Maybe we need a patch. And it can automate applying those patches. And Patch Manager can apply those patches for showing maintenance window for scheduling. But if it's such a thing that we determine, hey, this patch has to happen right now, that



can be triggered. In the Run Command lesson and also the automate lesson, we performed actions on our EC2 instance. And that's certainly the idea, but I didn't want to get too repetitive with the demo. So let's go out and do a demo. And in this demo, we're simply going to set up a patch configuration that we can then ultimately apply to our EC2 instance. We're not going to take it that far, but I want to show how this configuration is set up. All right, out at the AWS management console, I have two tabs open. It's a little tricky getting at some of these screens, and I just want to show two ways to do that. And I also want to show how to view the baseline patches that we have available to us. So let's click Quick Setup in Systems Manager. And this is a little bit surprising. We'll click Create, and there are a bunch of things that we can create. It's interesting to browse through and see what we can do from this page. But of course, we want Patch Manager. So let's click Create there. And from here, we can create our patch policy. But there's another way to get at this. Back to the Systems Manager main page, let's go to Patch Manager. Click that. And we can click Create patch policy, and there's our same screen. So two ways to get at that. I'll close this one and just leave this second tab open. We'll be using that. Back to quick setup. Let's give our configuration a name. All right. Now we can scan and find missing patches, or we can scan and install the patches that we need. So let's just do that. And we have a scanning schedule and an installation schedule and custom scan schedule. Well, we can set it up a little bit daily or automated with cron. I'll just use recommended defaults. Likewise with the installation schedule. Now reboot if needed. Let's keep in mind here, if we're rebooting someone's server, well, we have to keep in mind change processes. All right, patch baseline. We can use the recommended default or create a custom patch baseline. When I click that, there's all of our baseline patches for all of our different operating systems that we may have. Now there's a different way to get at this. Back to the other tab, in Patch Manager, click Start with an overview. And there is a tab, Patch baselines. And there's our same list of our default baselines. I believe it's ordered a little bit differently, but it is the same list, even Raspberry Pi operating system. All right, back to our quick setup, and I'm just going to use the recommended defaults. Now we can store our output logs in S3, and I'll browse and just find a bucket that I have set up for the course. And I'll browse and just find a bucket that I have set up for the course, devopspro, and I'll choose that bucket. All right, targets. And as you might expect, AWS Organizations is involved here. Choose the accounts and regions you want to deploy to. Notice the OUs. That's Organizations right there. We can go custom, or we can go current account. And I'm just going to do that. Current region or choose regions, I'll go current region. And choose how you want to target your instances. All managed nodes. We could use tags, we can create a resource group, or we can just do manual. And I'll do that. And there's the list of the instances that I have in this account in this region. And I'll just select one to deploy patches to. Rate control. This is like in CodeDeploy, percentage of nodes to deploy a patch to at the same time. So this is defaulting at 10%. Error threshold. So if we exceed that threshold, let's get out of this patching. There's a problem. We'll figure it out and



come back. All right, instance profiles. So, if you have instances that do not have profiles, the quick setup will create one and attach the appropriate policies and permissions. This instance profile created is attached only to instances that do not have an instance profile already. So I'll go ahead and click that, and this is sort of an abbreviated summary page. Go ahead and click Create. So it is working on it, and we will see updates to this patching operation on this page. And we have a surprise guest here. But by this time, probably not too surprising. On the other tab, I snuck over to CloudFormation. And sure enough, a stack set is being created for this operation. And if you think about it, we could be doing multiple accounts in multiple regions. Well, that kind of screams stack sets. So that has completed, and probably our patching configuration successfully updated. So it's working on the patching. And again, we just have the one instance. Looks like things are going well. A lot of green, no red. And it's working on configuration, created by state manager. So six successful out of eight, and the other two are pending. So with just a couple clicks, now, yes, we were only doing one EC2 instance. But we could be in a big organization, and all of the accounts in all of our regions would be listed right here in this process. Yes, it would take longer to complete. But the setup, even if we had a bunch of accounts in different regions, we just went through the quick setup. That's all the longer it would take to set up. So that's a look at automating patching using this quick setup. So we can automate the process of patching managed instances, both security and other types of updates. Key feature is patching obviously multiple instances at a time. We can patch operating systems and applications. Remember we can do on-premises. They just have to have that Systems Manager agent and some IAM permissions. We've seen these patch baselines. We can patch. These baselines include rules for auto approving patches within days of their release, as well as a list of approved and rejected patches. Remember, the Systems Manager maintenance window capability for scheduling our patches. And finally, and we just saw this, we can install patches individually or to large groups of instances. Again, remember the tagging and patching groups. So that's a look at Systems Manager Patch Manager. I'm going to end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

AWS Config

Hey, Cloud Gurus. Welcome. In this lesson, we're going to take a look at AWS Config. And Config is a service that allows us to assess, audit, and evaluate the configurations of our AWS resources. So Config is going to continuously monitor and record our resource configurations in AWS, and then we can set up our automations based on the evaluation of our configurations. So in this lesson, we'll take an overview look at AWS Config. We'll dive a little bit deeper. And then we'll solidify all that knowledge with a demo from the management console and wrap up with some exam tips. So with that said, let's check out AWS Config. So with AWS Config, we can really create an asset inventory in our AWS accounts. And



Config will allow us to record and assess the configurations. And how do we assess? Well, we could certainly go into the Config monitor and view some of the results, but we can also set up some of our well-known automations here. So Config is going to record all of the configuration data that runs through our system. And then, with Config, we can build rules to help us ensure compliance. Certainly a valuable tool when we're preparing for or reacting to audits. And as always, we can set up EventBridge to trigger our automations. Now I'm showing Config with EC2, and that's because of the demo. But Config can work with many different AWS services. With Config rules are things such as the detect drift on CloudFormation stacks. Our S3 buckets aren't public. Make sure our EC2 instances don't have public IP addresses in certain situations. Server snapshots should not be public. Root users should not have access keys. And those are all rules we can implement using AWS Config. Can we use AWS Config over multiple accounts and regions? And the answer is yes, absolutely. An aggregator is an AWS Config resource type that collects configuration and compliance data across multiple accounts and regions. So it's going to grab all of that data, even across regions and accounts, and aggregate it. So with all that said about AWS Config, let's go out to Config, implement a Config rule, and check out some other AWS Config rules. All right, so I'm out at the management console in AWS Config. Also have a tab open for EC2. I have two EC2 instances running, which will interact with AWS Config momentarily. So let's get started with AWS Config. I'll click View dashboard, and let's get our bearings a little bit. I'm going to go into settings. So recording, I already have on. Recording strategy. Record all resource types. Override settings. Notice a few things in IAM excluded from recording. And then just a bunch of things Config can record for us. Where is this information stored? In S3, config-bucket. Okay, so the settings look okay. Let's go over to Rules, and I have a few rules set. S3 bucket versioning enabled. And I have a bunch of non compliant resources. That's fine. S3 buckets without versioning enabled. And of course, if we want to use S3 as a repository, we would enable versioning. And certainly, if you're in that sort of shop where that's what you want to do, this rule could be helpful. We can set the desired instance type, and let's check that out. And it's showing a value of t3.micro. Now notice, this could be a comma-separated list of multiple types, but we're looking for all of our instances to be t3.micro. And we do have non compliance here. Note this ID number. Bounce over to EC2. There it is. And lo and behold, yes, it is not compliant. It's a t2.micro. Here's another rule, approved AMIs by tag. So in this rule, we want to check whether an instance is using a specific AMI. And we have some non compliance. And finally, EC2 instances, no public IP if the public IP field is present in the EC2 instance. So, note our ID. I'll go over to EC2, and there we go, public IP address. Whereas, check these other ones. No IP address. No IP address. And no IP address. So really, the scenario is if we really want to lock things down and get rid of those public IPs, we certainly have other methods of connecting to our instances. We don't need to have public IPs. All right, let's go back over to Config. And of course, we could add a rule, AWS managed rules. Create a custom Lambda rule. Create a custom rule using Guard. There are



392 AWS rules. So, you'll probably find what you need in this list. Let's back out of here. One more thing I wanted to show. So, this aggregator, AWS Config resource type that collects Config data from multiple accounts and regions. So we can go across accounts, across regions, and we can have an aggregator. Allow Config to replicate data from source accounts into an aggregator account. Must select this checkbox to continue. It's grayed out until we check that. So we can add accounts, and we can specify all of the regions, most likely specify the specific regions that we want information on. And then go ahead and create the aggregator. I'm not going to do that but certainly wanted to show this feature. I'll cancel out. Sorry, one more thing I wanted to point out in the aggregator. I'll just check that so it's easier to read. But, as you would probably expect, the aggregator works with AWS Organizations. So we could add individual counts, but if we check Add organization, all accounts in your organization will be added to this aggregator. Selecting this option automatically enables the integration between AWS Config and Organizations only when the management account is creating the aggregator. And in my case, I'm in the management account. So, probably pretty obvious if you think about it. But yeah, across accounts, across regions. Certainly, Organizations can come into play here. All right, let's talk about some exam tips. And one thing about the exam, AWS Config and CloudTrail are often paired together, maybe one is a distractor for the other. So let's make sure we understand some similarities but also some differences. Ah, similarities. Keyword audit. They're both involved in audits. They're both monitoring tools for AWS services. They can both track changes and store a history of changes. And they're both used for compliance and governance. Now, those are the similarities. Specific differences we need to know to differentiate the two. So which one reports on what has changed? AWS Config. Now that's a little vague. And certainly, CloudTrail is reporting on changes, but it's reporting on changes that— Well, let's go here. Spoiler alert. Which one focuses on AWS resources, not API calls? AWS Config. So AWS Config is focusing on changes to resources. Probably see where this is going. Which one reports on who made the change, when the change was made, and from where? CloudTrail. So you're getting all of this information. Which one focuses on events or API calls that initiate changes? That's CloudTrail. And really a keyword, key term. If you see API calls and you're talking about auditing, you want to be thinking CloudTrail. Now which one uses an aggregator to view data from multiple accounts, regions, and AWS Organizations? That's AWS Config. Now, just to clarify, you can certainly aggregate data from CloudTrail. But AWS Config specifically has an element in AWS Config called the aggregator. So be careful with that. So that's a look at AWS Config as it relates to policies and standards automation. And again, really make sure you understand the difference between Config and CloudTrail. So with that, I'm going to end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

Amazon Macie



Hey, Cloud Gurus. Welcome. In this lesson, we're going to talk about Amazon Macie. So what is Amazon Macie? Well, the official definition is it's a fully managed data security and data privacy service that uses machine learning and pattern matching to discover and protect your sensitive data in AWS. Now an unofficial definition and probably easy to remember, Amazon Macie analyzes and helps you protect data in your S3 buckets. Associate Macie with data in S3, and you'll be in good standing when you take the exam. So in this lesson, we'll take an overview look at Amazon Macie. We'll check out Macie in action. We'll go out to the management console and do a little demo with Macie. And then we'll come back for exam tips. Amazon Macie is one of those services which we'll compare and contrast with some other distractors that are often used for Macie on the exam. So let's check out Amazon Macie. All right, what is Amazon Macie? Security service that uses machine learning to automatically discover, classify, and protect sensitive data in AWS. Again, that's sort of the official version. But we're really looking to make that association with S3. How does it recognize data? Amazon Macie recognizes sensitive data such as personally identifiable information. Points that out to us. And we can take corrective action. Provides us with dashboards and alerts that gives visibility into how this data is being accessed. So let's check out Macie in action. So we have our AWS account. We enable Macie. And we'll see this when we go into the management console. And it's going to start analyzing our data. And again, that data is in S3 That is the big association you want to make, Macie with S3. And then it's going to provide us some feedback in a dashboard. So we enroll our AWS account with Macie. There's a 30-day free trial. Select our S3 buckets for content discovery and classification and review the accumulated alerts in the Macie dashboard. It's pretty darn simple. And those are really the key elements we need to know for Macie to prepare for the exam. And let's just reinforce that knowledge and go out and check it out from the management console. All right, out at the management console at AWS Macie. And some of this information on the main page is just useful in studying Macie. Discover our sensitive data using machine learning, continual evaluation of our S3 environment. And what types of things is it looking for? Unencrypted buckets, publicly accessible buckets, buckets shared outside our organization. So we'll just review some of this information. Down below are some use cases. Assess our data privacy and security across our entire S3 environment. So I'll go up here, and let's just get started with Macie. Notice, there's no charge for the first 30 days. So I'd probably encourage you to just watch the video rather than enabling Macie on your own accounts. This is something you could forget about, and then a month or two later start seeing charges on your account. When you enable Macie, Macie automatically creates a service link role. That's no surprise by now. And that gives Macie permission to perform tasks such as gather information about the data we store in S3 and evaluate and monitor our S3 buckets. So I'm going to go ahead and enable Macie, and it's already evaluating my S3 buckets. Another thing we can do is create and schedule our own jobs. So let's check that out real quick. And we can go ahead and analyze selected buckets. So I'm just going to analyze a recently



created bucket. Notice we can schedule jobs for this, daily, weekly, monthly. I'll just go daily. Or probably even a better option for this demo, a one-time job. All right, so we have some additional settings we could set. Enter criteria that determines which objects to include or exclude. If you don't enter any criteria, the job analyzes all objects in the bucket. So we'll stick with that. Click Next. Select managed identifiers. And it gives us a list here, recommended identifiers. Or we could go custom. And we can just refine this, AWS credentials, different countries. Actually, I'm just going to back out of this, and I'll just go with the recommended. Click Next. Haven't created any custom identifiers, and that's okay. Next. An analogy I'd make here, this is analogous to a whitelist. Allow list defines specific text or text patterns to ignore. So we're ignoring it. We're saying, hey, that's okay. Don't worry about that. Click Next. After all that, we just give our job a name, optional description. Click Next. Here is our review. Notice, estimated cost. Now if your 30-day trial was up, you'd see that cost estimation. And let's submit. And this is good. So you haven't configured a repository for sensitive data discovery results. Must configure this repository within 30 days of enabling Macie. Macie stores your sensitive data, discovery results, and findings for only 90 days. We could configure that now. But just doing this for the demo. I'm not going to be around using Macie for 30 days. So, let's just go back to the main page. Now remember, when we enabled Macie, it started analyzing our existing buckets. And we have some findings. Notice, 3 out of 48 of my buckets are publicly accessible, and that may be okay. I may have a public static website created from an S3 bucket, but certainly something to double-check. Am I doing encryption? Am I sharing any buckets? And the answer to that is no. So that's good. So that looks pretty good. Let's see about that individual job that we created. And that is still running. But realistically, there's not anything going on with that bucket. I know that bucket well, so that's going to be fine. I just wanted to show how to create that job. So we'll just end the demo here. Key takeaways and key words with Macie. Certainly, associate Macie with S3. And when you start thinking about security and sensitive information in S3, think Macie. And another key term, it's actually an acronym, PII, personally identifiable information, and that would be in our S3 bucket. You certainly want to start thinking about Macie when you're looking at your selections of answers. So I'm really just using Macie for this demo, and I don't want to forget about that free trial. Let's go over to Settings. We could suspend. You will not be charged for using Macie when the service is suspended. My situation, I really just use it for demos. Disable, stops monitoring and stops performing other activities in your account. Macie permanently deletes all of your existing findings, jobs, and resources. And that's fine for how I'm using it, so I'll disable. Just a little sanity check here. Type in Disable and click Disable. All right, let's check out some exam tips for Macie. And I'm going to focus on distractors for Macie. You'll see in scenario questions Macie mixed up with GuardDuty and Inspector, and we know what each of these is. Inspector is security assessments on our applications and those instances on which they run. GuardDuty analyzes the entirety of our AWS accounts. Macie protects our sensitive data in AWS. Where? Macie is focused on S3. So, keywords for Macie



again, S3 and maybe PII, personally identifiable information. So if you see those terms, start thinking Macie. So that's a look at Amazon Macie. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Secrets Manager

Hey, Cloud Gurus. Welcome. In this lesson, we're going to talk about AWS Secrets Manager. And Secrets Manager is an AWS service. Makes it easier for us to manage secrets such as database credentials, passwords, API keys. So, this probably sounds familiar. Doesn't that sound like Parameter Store? Well, we'll talk about the differences between the two. So in this lesson, we'll take an overview look at Secrets Manager. Check out the Secrets Manager workflow. And we'll do that comparison between Secrets Manager and Parameter Store. There are a couple key differences. And then we'll do a demo. We'll set up a secret and set up secret rotation. And then finally, we'll wrap up with some exam tips. So let's check out AWS Secrets Manager. All right, what is AWS Secrets Manager? Well, we're going to store our secrets and manage those secrets. Again, they can be database credentials, passwords, API keys, arbitrary text. What about access keys and secret access keys? Sure, we can store those as well. How does it work? We can store and control access to these secrets centrally. We can use the Secrets Manager console, which we'll do in this demo. We can use the CLI, or we can access secrets from our code with the API and SDKs. So Secrets Manager is going to interact with a lot of services, but here are some key services. KMS. We can encrypt our secrets, and we can create custom KMS keys or just allow AWS to create those keys for us. Can interact with CloudWatch, SNS, all of our messaging automations, CloudTrail, about what CloudTrail does. Alerts us of changes, who made the change and when they made the change. So obviously, if our secrets change, probably want to be aware of that. RDS. We're going to interact with an RDS database in the demo. And Lambda Functions. Let's talk about the workflow of working with secrets. So we have an administrator and an RDS database. The administrator creates credentials on that database, login credentials. But the next step the administrator takes is to store those credentials in Secrets Manager. So, our application through our code can query Secrets Manager for the secrets. And I'll reference back to our Parameter Store lesson. Remember, we pulled secrets out of a CloudFormation template into Parameter Store, and then we could reference those secrets in Parameter Store. Same same concept with Secrets Manager. So our application queries Secrets Manager for those secrets. Secrets Manager retrieves the secret, here's the key point, decrypts it and returns it to the application. The application now has those credentials for the database and uses them to access the database. That is the workflow. And again, very, very similar to Parameter Store. We'll talk about the differences momentarily. This is just a recap of what our secrets can be and how we access them. And in the past, we had to embed credentials for accessing the database directly in the application. That is very bad practice. In fairness to developers,



again, they created config files which would store these secrets and access them in a similar way to what we're doing here. But now, we have Secrets Manager. So we can replace hardcoded credentials with API calls to Secrets Manager to retrieve those secrets programmatically. We can also configure Secrets Manager to automatically rotate our secret according to a schedule we specify. All right. So this is almost identical to how we're using Parameter Store. Let's talk about some differences. Why do we even need Secrets Manager if we have Parameter Store? And along those lines, I'm not going to go to it. But here's a link to an article I wrote a couple of years ago comparing Secrets Manager with Parameter Store. But let's get to a couple of key points here. Cost. Parameter Store is free. Secrets Manager comes with a cost. Doesn't sound like a lot. But if you have a lot of secrets, it adds up. Secret rotation. Automatic secret rotation in Secrets Manager, big key difference between the two. Now with Parameter Store, you could set up secret rotation maybe with a Lambda function. But why would you when you have Secrets Manager? Another difference, cross-account access. And notice the secret size. You have a little bit more room, over double the size in Secrets Manager. This key difference is what you might be accountable for on the exam. Secret rotation automatically, that's with Secrets Manager. So let's just go out to the management console and set up a secret and set up secret rotation. All right, out at the management console in Secrets Manager, and we want to store a new secret. And I have created a database for this demo. So we'll store the username and password for this database. Reading here how it works, benefits and features, rotate secrets easily and safely. Let's go ahead and store a new secret. We have options here, credentials for RDS, which is what we're going to use. DocumentDB, Redshift, other databases, other types of secrets such as maybe an access key. But of course, we have a MySQL RDS database and their credentials. This is the username and password for the database. Notice, encryption key. You can encrypt using KMS or a customer-managed key. I'll just stick with the default. Let KMS do it. All right. So it's showing a database here because that's what we selected. This is the only available database in my account. I select it and click Next. Give our secret a name. We could add tags. Could add or edit a resource policy so that we can access secrets across our accounts. And look at this. Create a read-only replica of your secret in other regions. All right, let's talk about rotation. I'll click Next. Configure secret rotation. And again, this really differentiate Secrets Manager from Parameter Store. This automatic rotation, have to enable it. Notice, it's all grayed out. We just have to this slide button over. And we can build a schedule here based on hours, rotate. I'm going to say days. And 60 is pretty standard. I'll go with 60 days. Notice that it says it rotates immediately when the secret is stored, so we can expect that to happen when we complete this. Then the schedule will begin. Create a rotation function. And that function is a Lambda function. And there's a suggestion for a name. That looks okay to me. I'll go with that. Rotation strategy. Basically, it's single user versus multiple users, and I'll just keep it simple. Let's click Next. And as in much of AWS, we have an opportunity to review before we create. This is pretty cool. So, it's showing the code to access your



secrets from code in Java, JavaScript, C#, Python, Ruby, Go, and Rust. And then finally, we can click Store. One point about this I want to mention, and we will see this if we haven't seen it already. It's unfortunate it doesn't show it, but we can access our secrets from CloudFormation as well using either YAML or JSON. With that said, I'll click Store. Successfully stored the secret to show the list. Choose Refresh. As is often the case, CloudFormation is under the hood, CloudFormation is setting up rotation resources. This can take up to 2 minutes. And I click on View details, and we just have some tabs. Let's go back to our list. Here is our new secret. And you really get a lot of bang for your buck out of this. As you see, it's pretty easy to set up. But this is really the industry standard for storing secrets in AWS. The lesson with some exam tips. Do you not share secrets in code, CloudFormation templates, Lambda Functions, or anywhere similar. If it feels like a bad idea, it probably is. Know the key differences between Parameter Store and Secrets Manager. Parameter Store is free. There's a cost associated with Secrets Manager. The other key difference is secrets rotation with Secrets Manager. And let's talk about a scenario. Do we have insecure secrets in our code? If so, start thinking about Parameter Store and Secrets Manager, pulling those secrets out of the code. And we did that in the Parameter Store demo. Secrets Manager demo, I wanted to focus on secrets rotation. Do we need to rotate secrets? Well, if so, that's where Secrets Manager comes into play rather than Parameter Store. Sure, we could probably write a Lambda function to rotate secrets in Parameter Store, but why? Secrets Manager does it for us automatically. So that is a look at AWS Secrets Manager. And with that, I'm going to end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Service Catalog for Organizational Control

Hey, Cloud Gurus. Welcome. In this lesson, we're going to talk about Service Catalog for organizational control. So what is Service Catalog? Well, let's start by defining a problem. So we have a large organization, and our end users have access to the applications in our organization. Now we may have multiple different types of end users. We may have the very ambitious end user who downloads and tries to use everything. The other end of the spectrum, we may have users who are completely lost and don't know what to use or how to use it. And that's where Service Catalog comes into play. So in this lesson, we'll take an overview look at Service Catalog and really define what it is. And for the exam, that's what you have to know, what it is. We'll reinforce that with a demo, and then we'll come back and just review some exam tips. So let's go ahead and check out AWS Service Catalog. Now a good analogy to make here is think of just a catalog, catalog of your favorite products. Maybe a bicycle catalog or a clothing catalog. It's neatly organized. You know exactly where to go. It has categories. Within those categories are products. Think of Service Catalog as a catalog for our applications and other services within our organization. Sort of the end user perspective. From an administrative perspective, it helps us



organize, govern, think of permissions, and distribute application stacks. And in Service Catalog, those application stacks are called products. So let's check out catalog a little bit. All right, so we start out with portfolios. Portfolio is a collection of products. Think of portfolios as folders with configuration information determines who can use those products and how they can use them. And of course, within the portfolios, we have product service or application for end users. So again, it's this catalog format. And finally, we have end users, and that will get into IAM permissions. What can our end users see? Who sets this up? An administrator. And in IAM, there is a policy, Service Catalog admin full access. So that administrator will have the access to set up Service Catalog, our portfolios, and add products to those portfolios. And then we can address permissions for our users. Give those users access to what they need and restrict access to applications they don't need. On the DevOps Pro exam, Service Catalog is one of those services that you'll certainly see a few times in scenario questions, and it's just important to know exactly what Service Catalog does. A little bit more here for our end users. Too little access, too much. Do our end users have the correct level of access? And that's the admin's job. Keep everything updated and compliant with security standards. So there's definitely a compliance aspect to this as well. But again, we need to know what Service Catalog does and sort of understand its makeup. How are we distributing this? Portfolios, think of them as folders and then our products. Think of applications. The end users who use it, admins who configure all of this. So let's just reinforce these concepts. We'll go out to the management console and check out Service Catalog. All right, out at the management console in Service Catalog, and this is really good information here. Why Service Catalog, Ensure compliance with corporate standards, Centrally manage and distribute products. So this is admin functionality, but self-service discovery and launch. Browse and access products you want to launch on your own. That could be an end user. All right, to reinforce this, let's check out this Service Catalog. And it's probably not a surprise by now. But behind all of this is CloudFormation. All right, let's click Get started. And we have some options here. Getting Started Library provides a library of well-architected product templates so you can get started quickly. And we have the get started with an AWS CloudFormation product. So again, behind the scenes, CloudFormation. We download CloudFormation template. Create a key pair. Do you want a key pair or not? This is a good tip off that, hey, we're creating EC2 instances. We are creating a portfolio. Think of the portfolio as a folder of applications. All right, so if the portfolio is a folder of applications, think of the product as an application. Now the product doesn't necessarily have to be an application. But in most cases, it is. Step 5, add a template constraint to limit instance size. This is cost control. And as is most often the case, we need an IAM role assignment here. All right, so we've created the portfolio of products. Again, think of those as applications. And finally, we would grant our end users appropriate access to the portfolio. So if you're interested in Service Catalog, by all means, go through this demo or tutorial and doesn't take long. But the key points about Service Catalog is understand what it does and certainly



understand about portfolios, otherwise known as a folder and product. Doesn't have to be an application, but that's a good way to think of those. Most likely, it is an application. Well let's just jump ahead. I'm not going to do all of this, but let's look at a portfolio. All right, and this gives us a link to the management console Service Catalog. At the main page, I'll select portfolios, and we can create a portfolio. Just call it SCDemo, owner name, and Create. Go into the portfolio, and the next step would be Create product and then ultimately start adding products to the portfolio. And I'll just click Create product. I'm not going to go the whole way through this. Just want to ingrain some of these concepts. You can create a Service Catalog product based on multiple Infrastructure as Code languages, CloudFormation, Terraform, or external. Give the product a name, Owner. We're using a template to build our product. Cool stuff. Probably deeper than we need to go with it for the exam. So I'll just leave it at that. So again, just know what Service Catalog does and what it contains. Portfolios, think of it as a folder. And products, think of as applications within the folder. And our end users, ultimately, you want to give them access to the products applicable to their job. So for the exam, again, if you know what Service Catalog does and what it contains. Portfolios, think of them as folders. Products, think of them as applications. And then we just add some governance. Administrators add governance and permissions and restrain, basically giving our end users what they need to do their job. For the exam, if you just have a good idea of what's going on this diagram, you'll be well served for Service Catalog on the exam. All right, let's check out some exam tips. Components of Service Catalog, portfolios, product, and our end users. End user Valerie wants to spin up an EC2 instance. Can she? And this is sort of a trick question. Well, we don't have enough information to determine that. It depends on the permissions Service Catalog admin has given Valerie. All right. And as always, some of these sort of supplementary services on the exam, we're going to run into distractors. So let's test our knowledge. Organizations helps you centrally manage and govern your environment as you grow and scale your AWS resources. Sort of similar to Service Catalog, but Service Catalog is focused on those products, which again are mostly applications. We know what CloudFormation is. Lets you model, provision, and manage AWS third-party resources, Infrastructure as Code. Trusted Advisor. Keywords for Trusted Advisor recommendations and best practices. So that is a look at AWS Service Catalog. And with that, I'm going to end this lesson. Thank you for joining me, Cloud Gurus. I will see you in the next lesson.

Security and Compliance Review

Supporting Services for Security and Compliance

Hey, Cloud Gurus. Welcome to this lesson, Supporting Services for Security and Compliance. So in the lesson, we'll take a look at some of these services in



the domains for security and compliance. They may not be prominent figures, but we should at least know about them and what they can do for us. So we'll check out AWS CloudHSM, AWS Directory Service for Microsoft Active Directory, and we'll check out Resource Access Manager, and then finish up as always with some exam tips. So let's start talking about CloudHSM. And this is one of these services where most people have heard of it, but maybe not too many have worked with it. So what is AWS CloudHSM? It is an enterprise-class service used for secure encryption key management and storage. Now the really key point about this is that the CloudHSM, it's an actual hardware device that you own. It's customer-owned, and it's also single-tenant. So it's all yours. So taking a look at the diagram, we have our AWS resources just showing an EC2 instance application running on it and also an HSM client. And we interact and we get our keys. Again, that ownership and single tenancy is the key point. That might come down to compliance requirements. So CloudHSM, hardware security module in the AWS cloud. It is a computing device that processes cryptographic operations and provides secure storage for our keys. So, let's just talk about the shared responsibility model for a second. Who manages the hardware device? Now remember, we own it. It's customer-owned. AWS manages it. Who manages the encryption keys? You do, the owner of the device. So maybe looking at encryption from a hierarchical perspective, let's start at the bottom. So, you don't need to manage the keys. You just need encryption. What's your solution? Well, there's default encryption, and that'll work just fine for that use case. Now, what if you need to manage only the keys? In other words, you don't need to own the device. You don't need single tenancy. You just need to manage the keys. You can use Key Management Service and the customer-managed keys within Key Management Service. And then, you need to manage the keys and have dedicated hardware that's single tenancy, and that would be a key term for CloudHSM. If you see that single tenancy or maybe customer-owned, you want to start thinking about CloudHSM. So those are some key points on CloudHSM. Certainly, if we know this hierarchy and a couple of those key points, we'll be just fine with CloudHSM. All right, let's talk about AWS Directory Service for Microsoft Active Directory. And this has sort of an aka otherwise known as AWS Managed Microsoft AD. So let's check this out. All right, so let's say we have an on-premises site using Microsoft Active Directory. And of course, we have an AWS account or accounts, and we create a trust relationship and just showing a VPN as a connection. AWS Directory Service makes it easy to set up and run directories in the AWS cloud. So we can use our Microsoft AD in the cloud and/or connect our AWS resources with an existing on-premises Microsoft Active Directory, also known as AWS Managed Microsoft AD. So if we just know these couple names and sort of this alias, AWS Managed Microsoft AD, probably won't get tripped up with this. And just understand what it's doing for us, sort of bringing our Microsoft AD into the cloud. All right, let's talk about AWS Resource Access Manager. So, a big focus in this exam update is working with multi-account environments, and Resource Access Manager helps you securely share your resource across AWS accounts within your organization or organizational



units. So, let's say we have an account, account A. We've set up all of our resources. Everything's working great. Now guess what? We have other accounts that want to use these resources. So, using AWS Resource Access Manager, and you'll see the acronym RAM. And certainly, in this instance, don't think of RAM memory. Think of Resource Access Manager. So the steps. We select the resources to share within this account A. We choose the appropriate permissions. Think about IAM roles and, of course, the underlying policies. And then we specify the relevant users. So it would be IAM users or organizations, or it could be IAM groups in these other accounts to give access to these resources. Now this could be a cool demo or certainly something to play around with on your own. But for the sake of the DevOps Pro exam, this is probably enough information. Sort of have to ration our time wisely in preparing for the exam. All right, with all that said, let's take a look at some exam tips. All right, so you've received compliance requirements that require you to have a dedicated hardware device and manage the encryption keys. Certainly the key term to pick out of the scenario would be dedicated hardware device, and that just screams CloudHSM. What is another name for AWS Directory Service for Microsoft Active Directory? AWS Managed Microsoft AD. And in a multi-account organization, how can you share resources across accounts? AWS Resource Access Manager. All right, with that, I will end this lesson. Thank you for joining me, Cloud Gurus. I'll see you in the next lesson.

DevOps Conversations for Security and Compliance

Hey, Cloud Gurus. Welcome to this lesson, DevOps Conversations for Security and Compliance. And in this conversation, we're going to listen in on a conversation between our DevOps lead and our junior DevOps engineer. So let's jump right into this conversation. All right, the DevOps lead says, I like the way you're progressing. We might need some help on the security team. Let's talk about security and compliance. DevOps lead goes on, we have this banking application using EC2 instances and an Auto Scaling group behind an Application Load Balancer. Additionally, customers fill out forms in the application, and the forms are stored in S3. This needs to be really secure. What would you use to detect PII information, personal identifiable information, in S3? What steps would you take to initiate automated remediation? So think about this scenario, and what are the key terms in this scenario? Well, certainly PII information. So what service in AWS is going to help us with that? And certainly, that would be a key term in a scenario question on the exam. If you see PII information, probably associate it with the service that we need. This could extend, as it should, to on the job. We may be sitting in a requirements meeting with our client. And this issue comes up, client needs to detect PII information in S3, and we can show the client that, hey, we're on top of this situation. We know exactly what you need. Let's hear what the DevOps engineer says, and we'll find out what we need. The S3 buckets can be scanned using Amazon Macie to



detect PII information. Findings can be sent to EventBridge and/or Security Hub to initiate remediation. Couple key points there again, most importantly that relationship, PII information. That is screaming Macie. In an S3 bucket for sure, think Macie. So, on the exam scenario question, start scanning your answers for Macie. On the job in a requirements meeting with your client, they talk about PII information in S3, well, go ahead and be the hero and say no problem. We can use Amazon Macie to detect that PII information. And our steps to initiate automated remediation. EventBridge, often a trigger for automations, and Security Hub. All right, let's continue. DevOps lead says, speaking of Security Hub, we want to enable Security Hub in this account. What else do we need to get started with Security Hub? This is a prerequisite for Security Hub. Think about it. Let's see what our junior engineer comes up with. To enable Security Hub, you have to enable AWS Config first. They are that tightly coupled. So just remember that. DevOps lead goes on, I'd like to receive automated notifications on Security Hub findings. How would you set this up? So highlighted in blue, sort of our key term here, automated notifications. I'm thinking of two services in AWS. Let's see if I'm on the same page with the junior engineer. And yes, the junior engineer might not be a junior for too long. We can use a EventBridge and connect it to an SNS topic that we create for Security Hub findings. Then we just need to subscribe your email, that being the DevOps lead, to the SNS topic. And maybe the key point from this little conversation is that a lot of these automations are reusable across different services. So if we want to automate a notification, EventBridge and SNS, that pattern will often be part of the solution. Now if we want to do a little bit more than just notify, maybe it could be EventBridge triggering a Lambda function, which can perform a remediation. But just know those automations. That'll go a long way for you on the exam and on the job. DevOps lead goes on. In some cases, we want to remediate findings. Are there any services that can help with this? Well, I actually just mentioned one. So think about other services that can remediate issues for us. Let's see what our junior engineer has to say. Well, that's the one I gave away, the first one. Sure, we can use EventBridge to trigger a Lambda function that can be written to remediate findings. Another tool is the Systems Manager Run Command to remediate issues on EC2 instances. An example being let's say that we got a finding in Security Hub related to a port being wide open to the world, 0.0.0.0/0, we could have an automation setup with the Run Command to go ahead and change that, maybe to an IP range that is suitable for our organization. DevOps lead goes on. What's a typical finding on EC2 that we should be proactive about addressing? So, think about when you launch an EC2 instance from the management console and think about some of the defaults that you are given. And there's one that just jumps out. It produces a Security Hub finding. Well, let's see what our junior engineer is thinking on this. Ah, there we go. By default, the SSH port is wide open. Since we're creating instances using CloudFormation, we should provide an IP address range instead. Some great answers from our junior engineer. DevOps lead is not convinced yet. More questions. We have outside contractors who occasionally need access to our



application. How would you handle this? Maybe a key term we want to pick out in a scenario question or from our client requirements, outside contractors, outside of our organization, occasional access. Let's think about that a bit. Let's hear what our junior engineer is thinking. I'd use roles and identity federation. I'd see how long they needed access, and I'd use STS to set them up with short-term credentials. So, outside contractors, outside of our organization, we're not giving them logins. Going to use identity federation and STS, short-term credentials. Is our DevOps lead convinced yet? Wow. This is a bit of a grilling. But our junior engineer is hanging in there great. Speaking of roles, we're spinning up a lot of dev and test environments and want to limit access to our prod environment. As you know, we have teams in the US, Asia, and Australia. What are your thoughts on least privilege access to these environments in a way that is highly scalable. So we have a bit of a paragraph here, but this paints a picture for us. So, least privilege or maybe thinking IAM here, highly scalable. US, Asia, Australia, certainly multiple regions, most likely multiple accounts. So sometimes we have to get that picture in our mind. We don't have time on the exam to sketch everything out. We have to visualize some of these. Just think about that. Maybe sketch it out. Let's see what our engineer comes up with. Well, it's a broad question. And this might actually be a good way to answer a question from a client. Don't sort of paint yourself into a corner being super specific with this. It's a broad question, but I think about using attribute-based access control instead of role-based. Now that ties to highly scalable, and attribute-based access is really good with that. Junior engineer goes on. We can tag resources based on the environment they're in and attach tags to users as well. This scales better for large geographically dispersed organizations. So this is sort of a use case for attribute-based access. And the question from the DevOps lead gave us that use case, highly scalable, multiple accounts, multiple regions, a complex access situation. And that's where attribute-based access really shines. All right, so our junior engineer is crushing it. Aha, DevOps lead is hard to impress, but I think we got there. I'm impressed with your knowledge. Touch base with the security team lead. I think you're ready for this. One would hope. All right, so that's a look at another DevOps conversation. Just get your mind thinking in terms of key terms and phrases and knowing the fundamentals of services. That'll certainly help with scenario questions on the exam and certainly on the job, maybe in conversations about requirements. And with that, I will end this lesson. Thanks for joining me, Cloud Gurus. I'll see you in the next lesson.

Policies and Standards Automation Scenarios

Hey, Cloud Gurus. Welcome to this lesson, Policies and Standards Automation Scenarios. So this is our end-of-section review and look at scenarios that can be helpful in exam prep. So we'll touch on our scenario prep real quick, how we want to approach this lesson. And then we'll have four scenarios. And if you follow the course in sequence, you've seen this before. So, let's just dive right in. Remember



our scenario prep. We want to read the scenario, try to visualize it, also maybe sketch it out on a notepad just to enforce that visualization. And then we'll check out the solution. And as always, remember, on an exam, we only have roughly 2 minutes per question. So we don't have time to sketch out every question. I save my sketching for the hard questions that I mark for review and save for the end of the exam. And almost always, by the time I get to these review questions, I've marked. Near the end of the exam, they're easier. It sort of gives my mind a little bit of time to absorb the content. And oftentimes, you'll run into other questions on the exam, which kind of jog your memory on some things. All right, with that said, let's check out some scenarios. We need to be notified if the number of instances in our account is reaching service limits. All right. So visualize that. Think about maybe the AWS service that can help with that. Sketch it out. Pause this lesson if necessary. But let's check out the solution. All right, so we have an Auto Scaling group containing our instances. And don't worry if you didn't visualize the Auto Scaling group. It could be just some lone instances. But what we want to do is set up a CloudWatch alarm. Ah. So, the service that's going to notify us of service limits is Trusted Advisor, in this case, coupled with CloudWatch. So we set an alarm on this metric, service limit usage. And that is a Trusted Advisor metric. That meets our scenario requirement. Let's check out another scenario. Instructions from an audit dictate that we ensure the instances in our Auto Scaling groups have up-to-date patches installed and the correct application version. What steps can we take? So again, visualize the scenario. Think about the AWS service that is going to help us with that. Maybe sketch it out. Pause this if you need to. But let's see the solution. Ah, Systems Manager inventory. It's going to inventory the instances in our Auto Scaling group. And Systems Manager inventory checks on these things, such as proper versions and patching. Now, the next step, taking this scenario a bit further, would be to set up our automation, maybe a notification automation, or take steps to correct this problem. Working with Patch Manager, we could get new patches out to these instances. So this is sort of a simplified scenario with some extra credit if you've thought about the next steps to take. Let's look at another scenario. You've joined a new company as an administrator and have been correcting security issues. Several files have been identified with plaintext secrets, access keys, and passwords. How can you secure this information? So we have our secrets, and we need to secure this information. So visualize that. Think about the services that can help. Go ahead and pause if you need to. Let's check out the solution. Ah, pull those secrets out of our files, out of our code, and put them in Parameter Store. And we can encrypt the data. All of these parameters can be encrypted and stored in Parameter Store. Now, we've gotten new requirements. We need to implement automatic password rotation for database passwords. How can we do that? And this is sort of a key term that should trigger thoughts of a specific service, and that service is Secrets Manager. Secrets Manager comes with a cost; Parameter Store is free. One of the big selling points is password rotation automatically. All right, let's check out one more scenario. We have multiple AWS accounts in multiple regions. We need to collect configuration



information on all of our resources. How can we do this? So try to visualize that. Pick out some keywords. All right, configuration information seemed like a key term here. Sketch it out. Go ahead and pause if necessary. But, on with the solution. All right, multiple accounts and these are across multiple regions. We need configuration information. Well, we can do this with AWS Config. Now, to take it a step further, how about we store this all in one place. We can set up an aggregator on AWS Config, gather this data across multiple accounts and regions, and what's not showing here is ultimately, we can store in an S3 bucket. All right, so those are some scenarios related to policies and standards automation. the AWS DevOps domain, and also end of our section on policies and standards. So congratulations, Cloud Gurus. I'll see you in the next section.